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(54) **JOINT PHASE AND GAIN CALIBRATION
FOR MILLIMETER WAVE BEAMFORMING**

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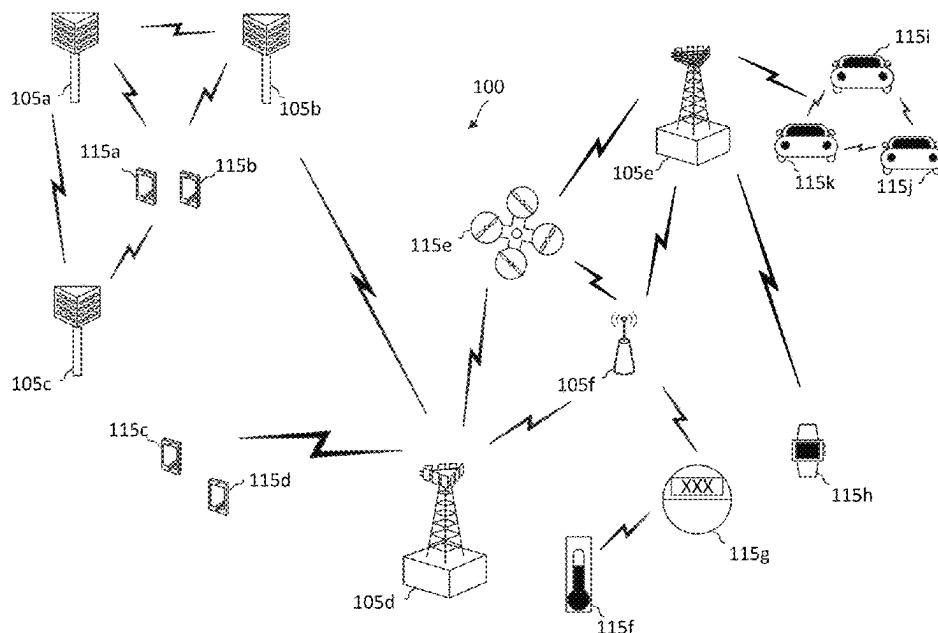
CPC H04B 17/11; H04B 17/21; H04B 17/12; H04B 17/14; H04L 5/0055

See application file for complete search history.

(57) **ABSTRACT**

Wireless communications systems, apparatuses, and methods are provided. A method of wireless communication performed by a user equipment (UE) includes transmitting, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions, receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array, and transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase/gain values and calibration coefficients associated with a second set of phase/gain values, wherein the calibration coefficients associated with the second set of phase/gain values are interpolated from the calibration coefficients associated with the first set of phase/gain values.

30 Claims, 10 Drawing Sheets



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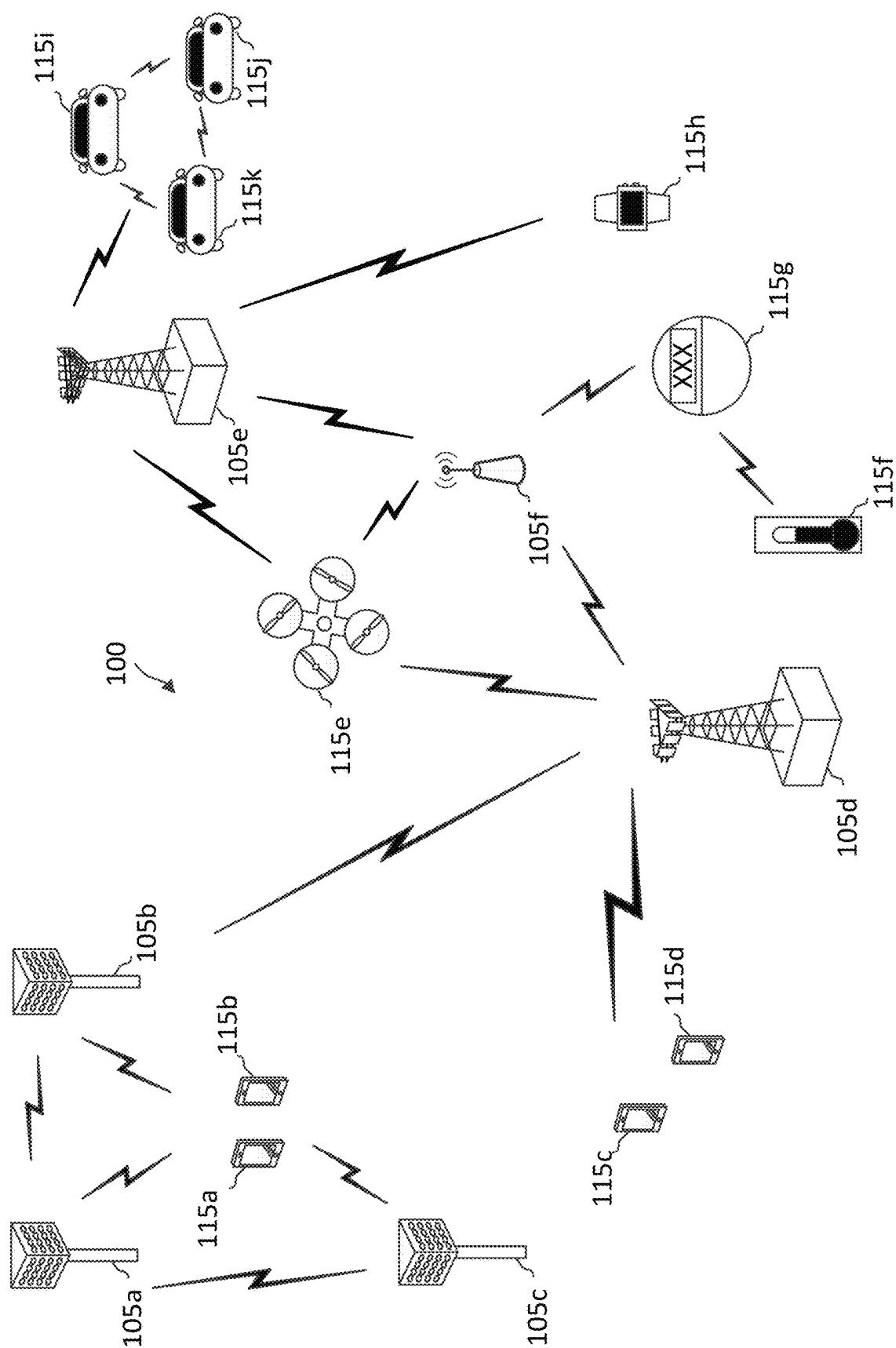
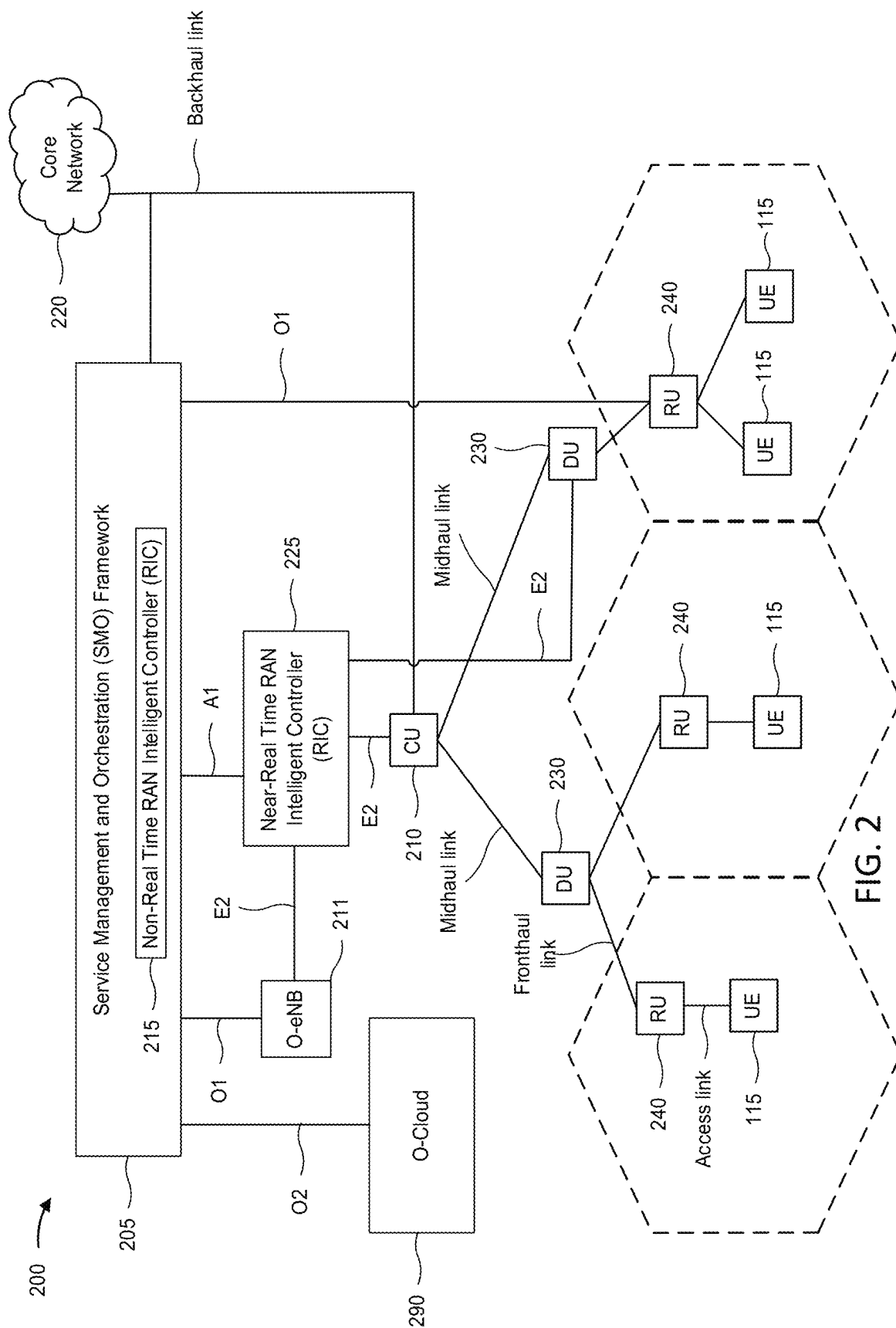


FIG. 1



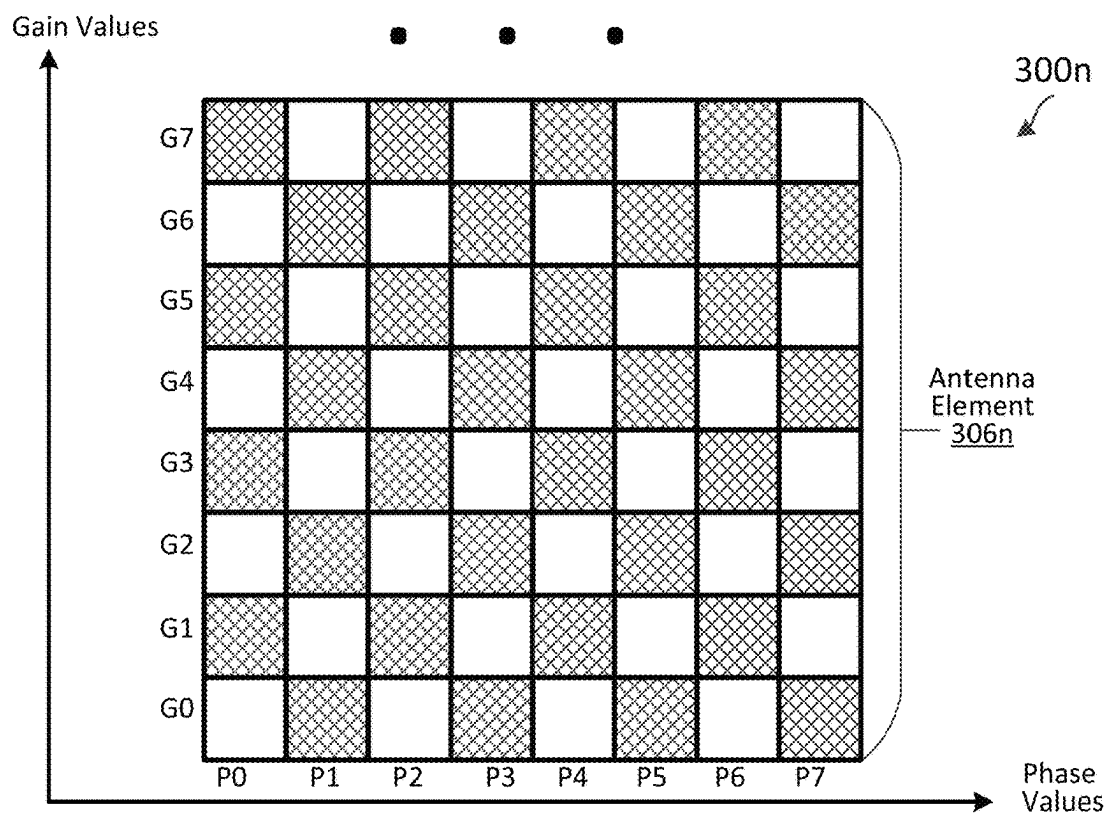
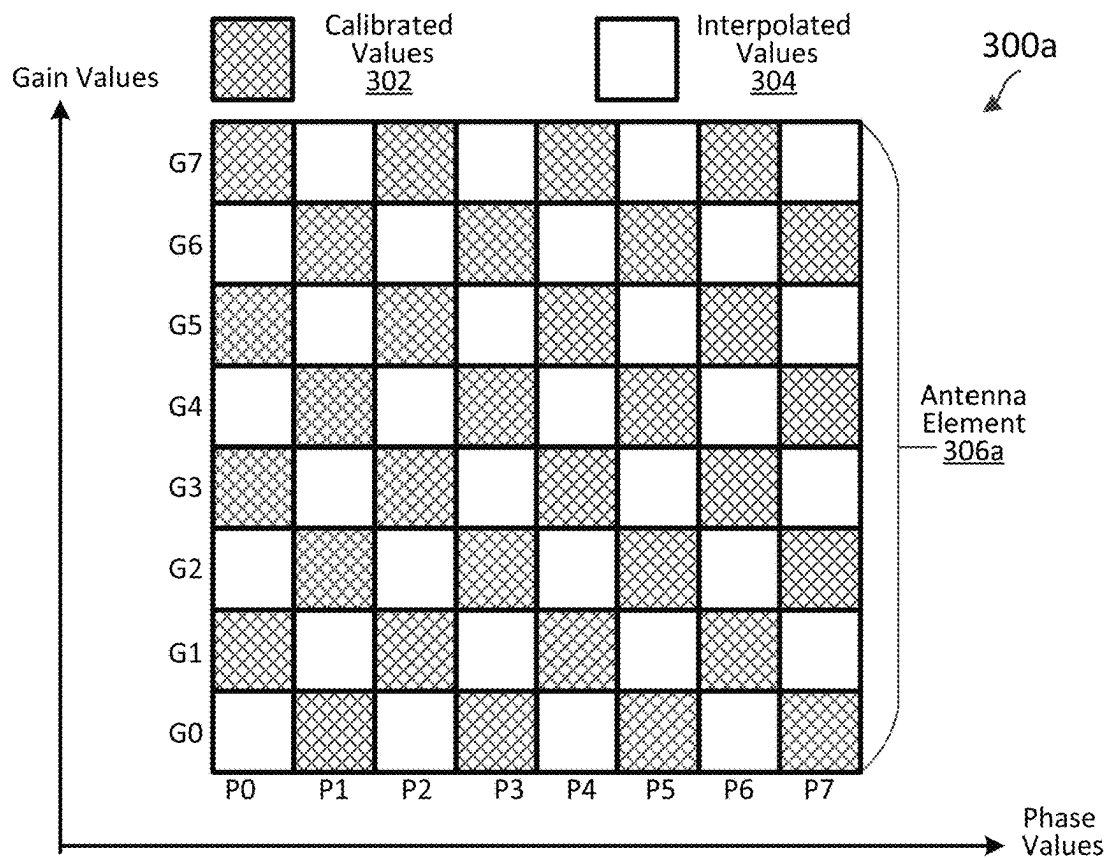


FIG. 3

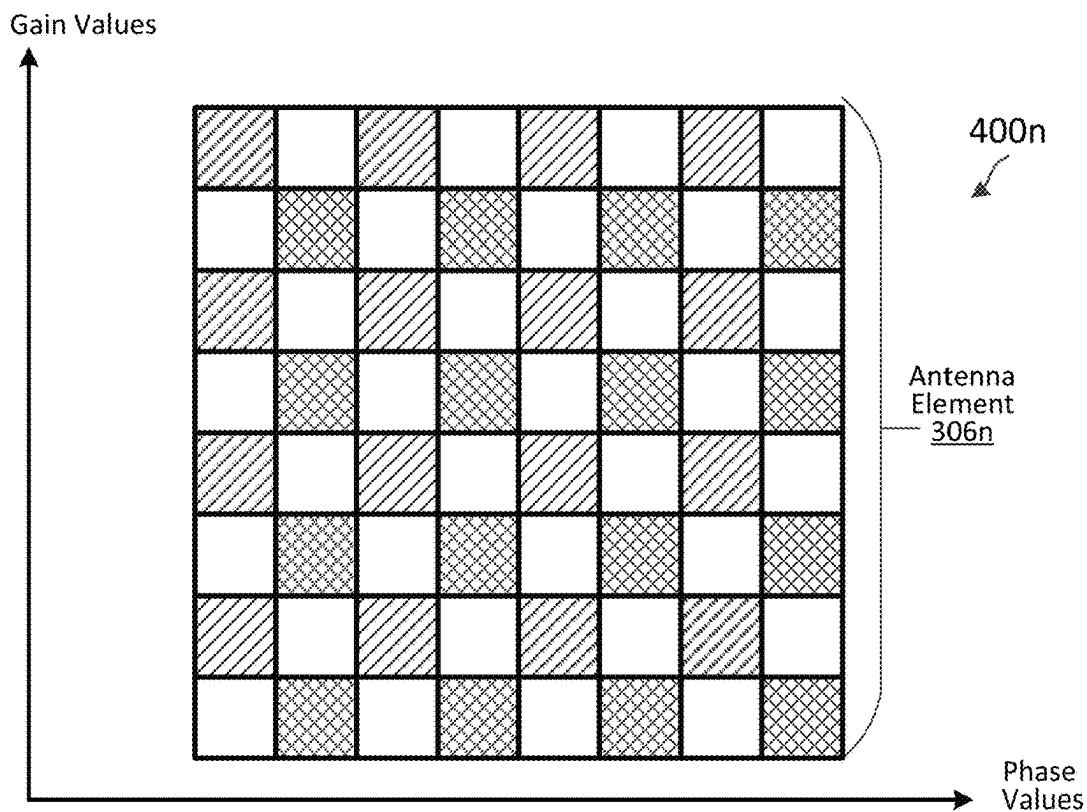
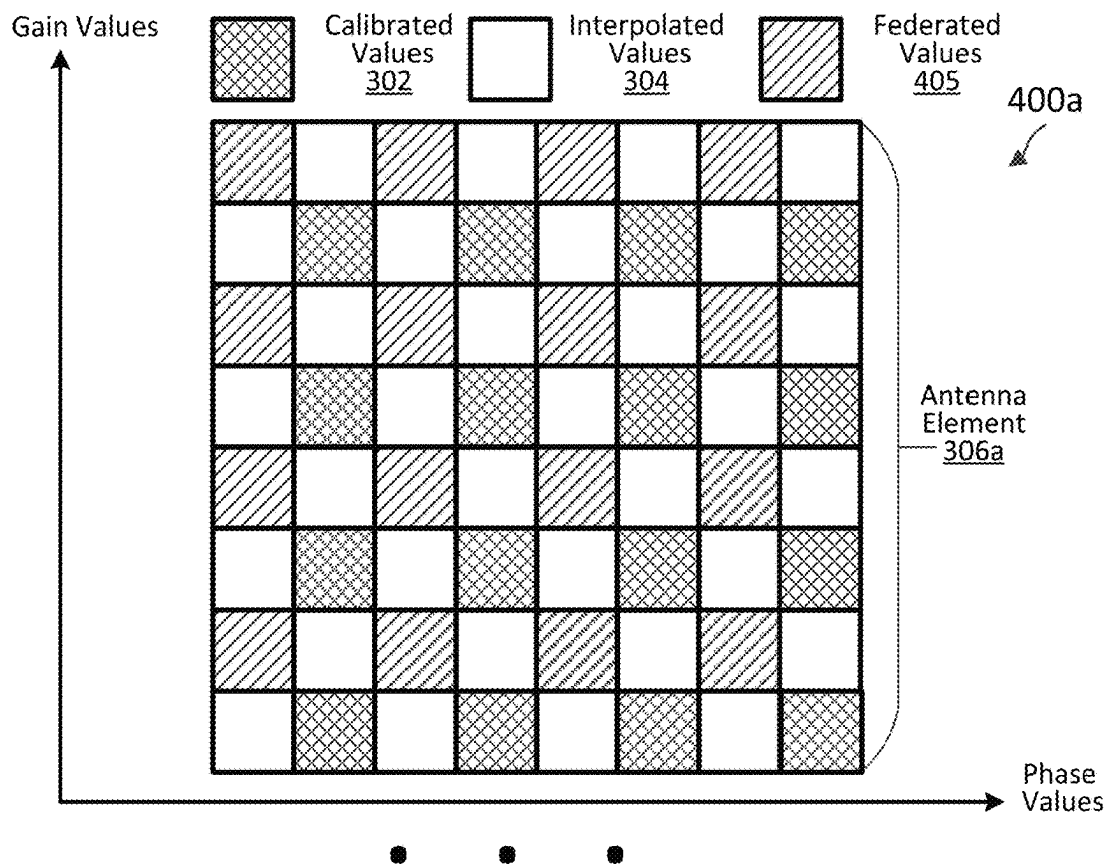


FIG. 4

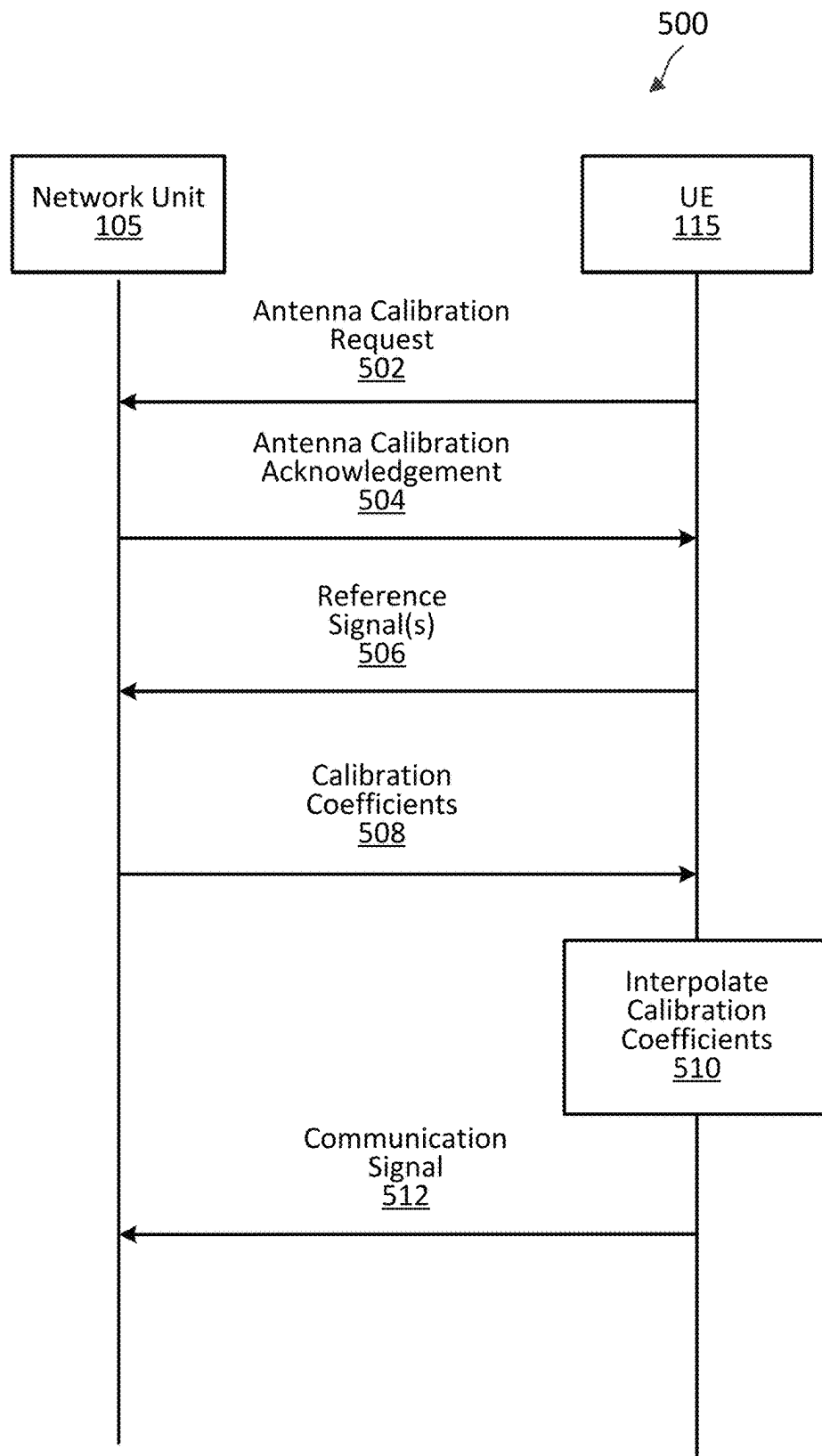


FIG. 5

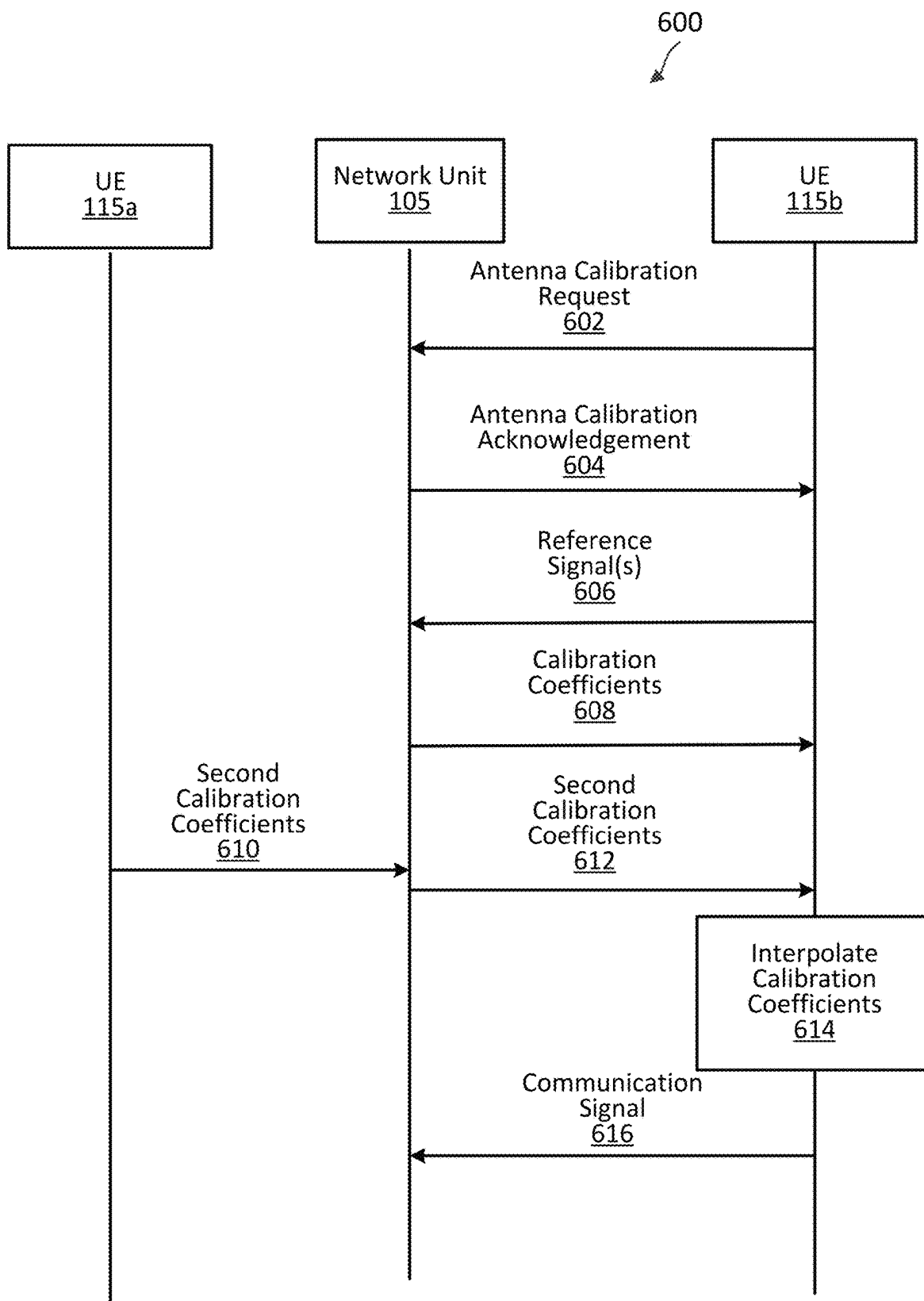


FIG. 6

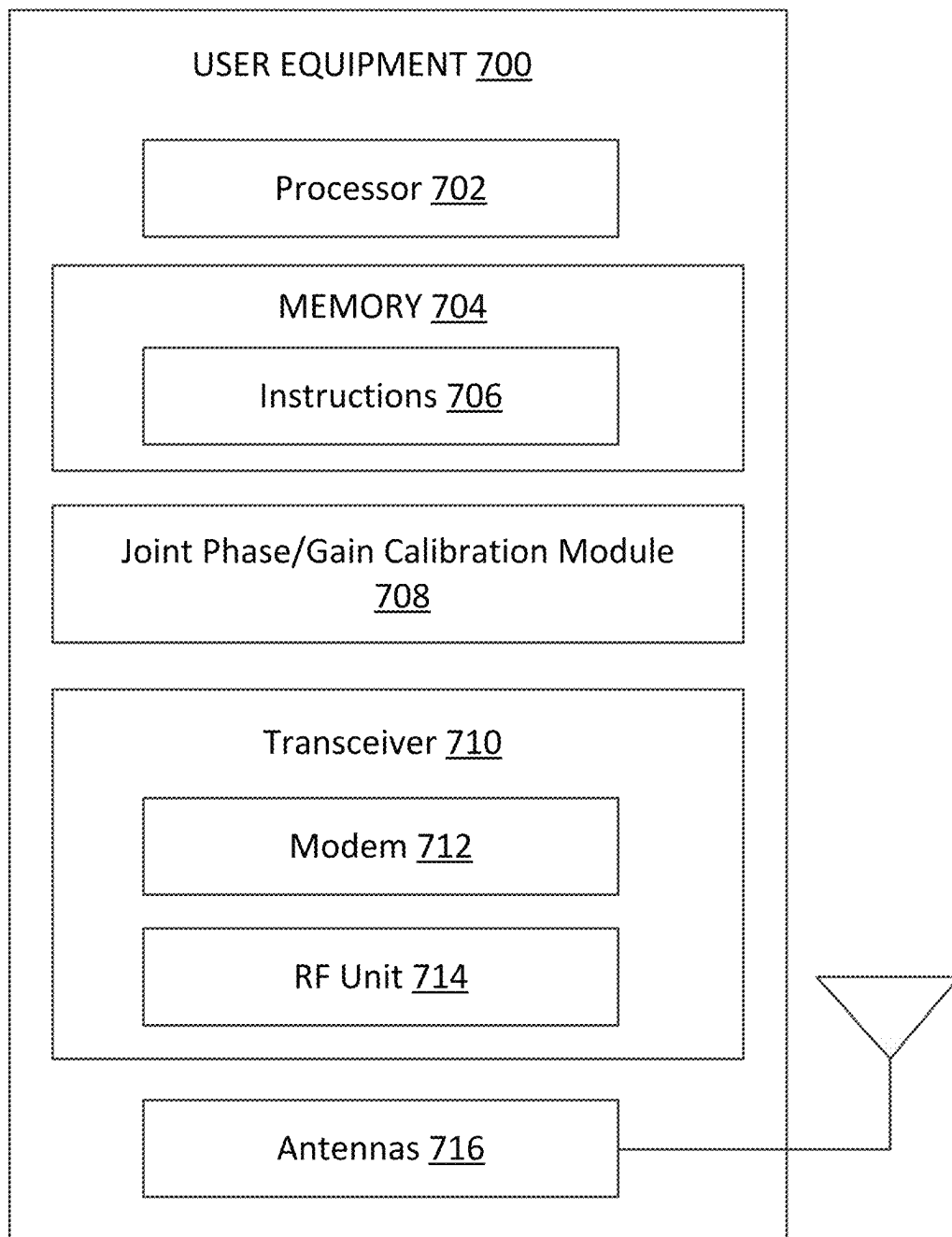


FIG. 7

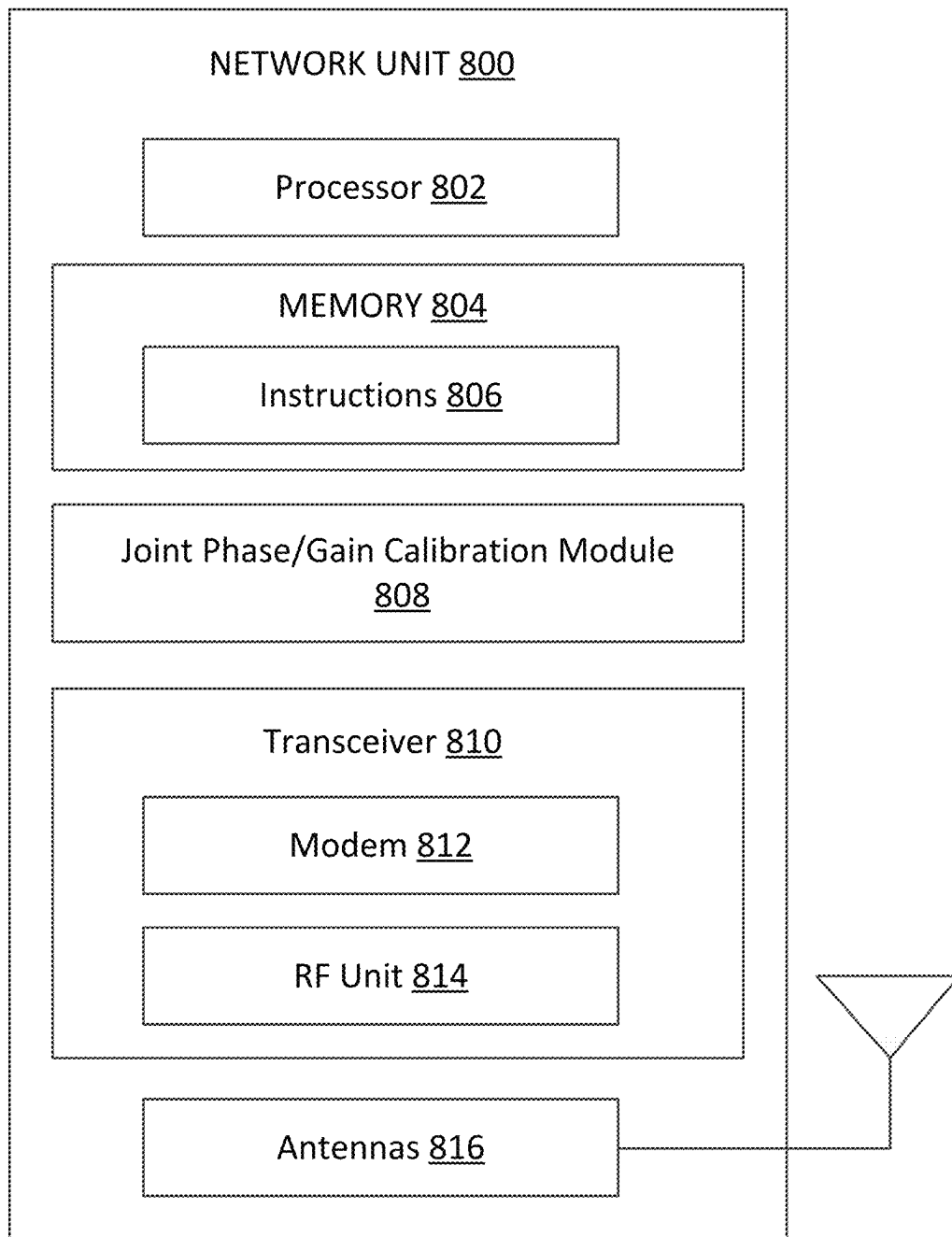


FIG. 8

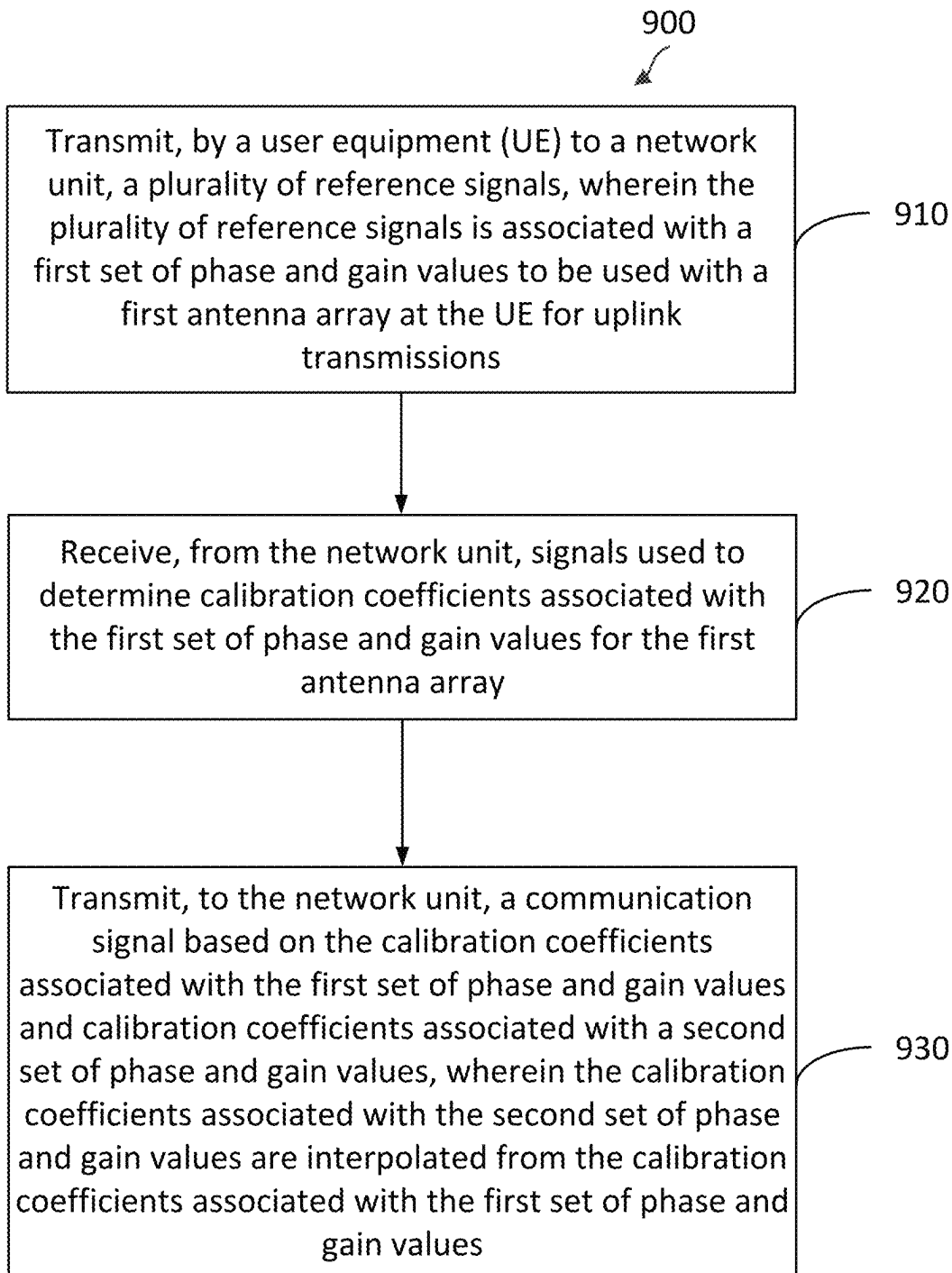


FIG. 9

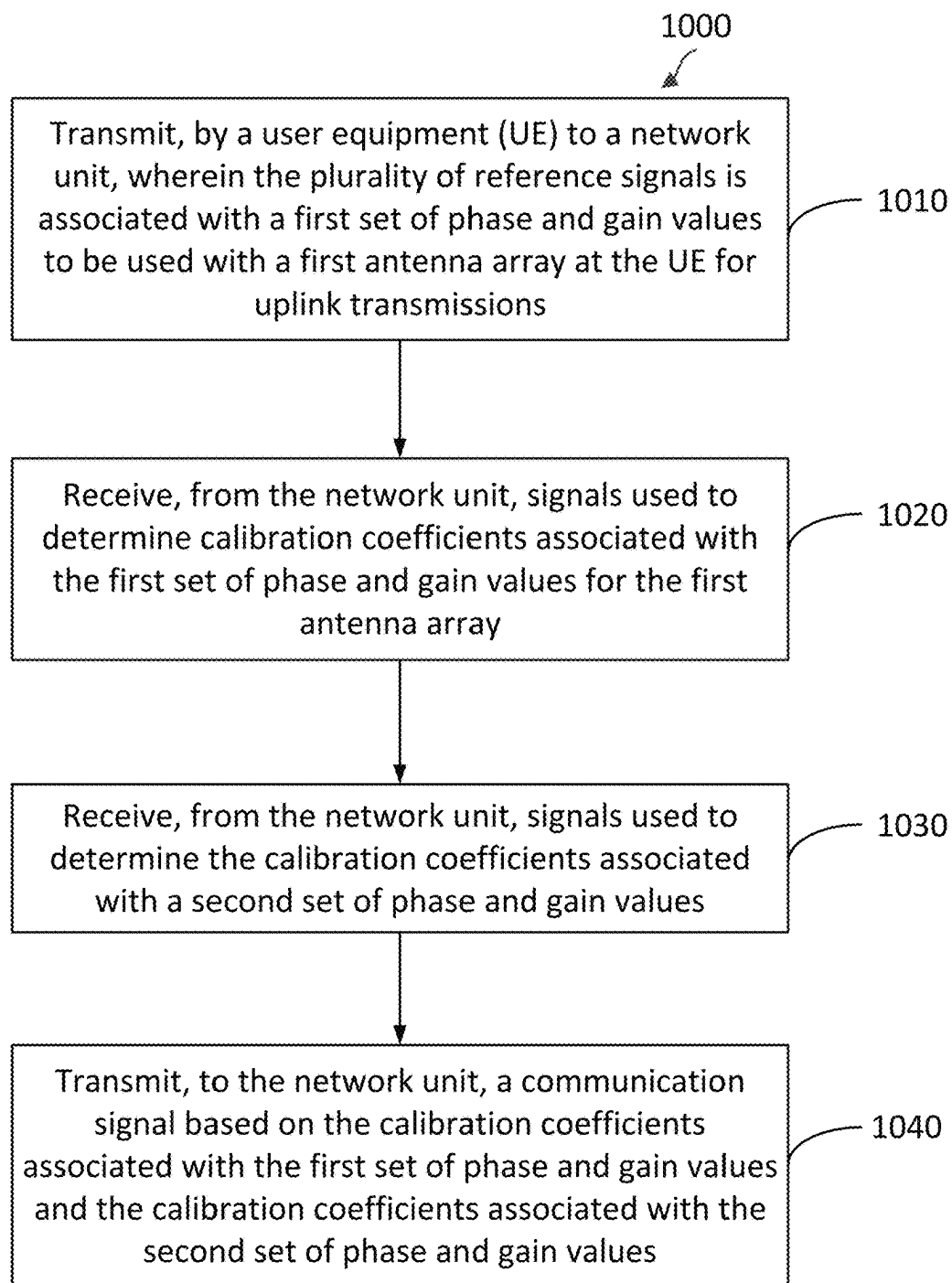


FIG. 10

JOINT PHASE AND GAIN CALIBRATION FOR MILLIMETER WAVE BEAMFORMING

TECHNICAL FIELD

This application relates to wireless communication systems, and more particularly, to joint phase and gain calibration for millimeter wave beamforming in wireless communication systems.

INTRODUCTION

Wireless communications systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). A wireless multiple-access communications system may include a number of base stations (BSs), each simultaneously supporting communications for multiple communication devices, which may be otherwise known as user equipment (UE).

To meet the growing demands for expanded mobile broadband connectivity, wireless communication technologies are advancing from the LTE technology to a next generation new radio (NR) technology. For example, NR is designed to provide a lower latency, a higher bandwidth or throughput, and a higher reliability than LTE. NR is designed to operate over a wide array of spectrum bands, for example, from low-frequency bands below about 1 gigahertz (GHz) and mid-frequency bands from about 1 GHz to about 6 GHz, to high-frequency bands such as millimeter wave (mmWave) bands. NR is also designed to operate across different spectrum types, from licensed spectrum to unlicensed and shared spectrum. Spectrum sharing enables operators to opportunistically aggregate spectrums to dynamically support high-bandwidth services. Spectrum sharing may extend the benefit of NR technologies to operating entities that may not have access to a licensed spectrum.

NR may support various deployment scenarios to benefit from the various spectrums in different frequency ranges, licensed and/or unlicensed, and/or coexistence of the LTE and NR technologies. For example, NR may be deployed in a standalone NR mode over a licensed and/or an unlicensed band or in a dual connectivity mode with various combinations of NR and LTE over licensed and/or unlicensed bands.

In a wireless communication network, a BS may communicate with a UE in an uplink direction and a downlink direction. Sidelink was introduced in LTE to allow a UE to send data to another UE (e.g., from one vehicle to another vehicle) without tunneling through the BS and/or an associated core network. The LTE sidelink technology has been extended to provision for device-to-device (D2D) communications, vehicle-to-everything (V2X) communications, and/or cellular vehicle-to-everything (C-V2X) communications. Similarly, NR may be extended to support sidelink communications, D2D communications, V2X communications, and/or C-V2X over licensed frequency bands and/or unlicensed frequency bands (e.g., shared frequency bands).

BRIEF SUMMARY OF SOME EXAMPLES

The following summarizes some aspects of the present disclosure to provide a basic understanding of the discussed technology. This summary is not an extensive overview of all contemplated features of the disclosure and is intended

neither to identify key or critical elements of all aspects of the disclosure nor to delineate the scope of any or all aspects of the disclosure. Its sole purpose is to present some concepts of one or more aspects of the disclosure in summary form as a prelude to the more detailed description that is presented later.

In an aspect of the disclosure, a method of wireless communication performed by a user equipment (UE) may include transmitting, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions; receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; and transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

In an additional aspect of the disclosure, a method of wireless communication performed by a user equipment (UE) may include transmitting, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions; receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; receiving, from the network unit, signals used to determine the calibration coefficients associated with a second set of phase and gain values; and transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values.

In an additional aspect of the disclosure, a user equipment (UE) may include a memory; a transceiver; and at least one processor coupled to the memory and the transceiver, wherein the UE is configured to: transmit, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions; receive, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; and transmit, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

In an additional aspect of the disclosure, a user equipment (UE) may include a memory; a transceiver; and at least one processor coupled to the memory and the transceiver, wherein the UE is configured to: transmit, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions; receive, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; receive, from the network unit, signals used to deter-

mine the calibration coefficients associated with a second set of phase and gain values; and transmit, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values

Other aspects, features, and instances of the present invention will become apparent to those of ordinary skill in the art, upon reviewing the following description of specific, exemplary instances of the present invention in conjunction with the accompanying figures. While features of the present invention may be discussed relative to certain aspects and figures below, all instances of the present invention may include one or more of the advantageous features discussed herein. In other words, while one or more instances may be discussed as having certain advantageous features, one or more of such features may also be used in accordance with the various instances of the invention discussed herein. In similar fashion, while exemplary aspects may be discussed below as device, system, or method instances it should be understood that such exemplary instances may be implemented in various devices, systems, and methods.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a wireless communication network according to some aspects of the present disclosure.

FIG. 2 illustrates an example disaggregated base station architecture according to some aspects of the present disclosure.

FIG. 3 illustrates joint phase/gain calibration values in wireless communications according to some aspects of the present disclosure.

FIG. 4 illustrates joint phase/gain calibration values in wireless communications according to some aspects of the present disclosure.

FIG. 5 is a signal flow diagram of a communication method according to some aspects of the present disclosure.

FIG. 6 is a signal flow diagram of a communication method according to some aspects of the present disclosure.

FIG. 7 is a block diagram of an exemplary user equipment (UE) according to some aspects of the present disclosure.

FIG. 8 is a block diagram of an exemplary network unit according to some aspects of the present disclosure.

FIG. 9 is a flow diagram of a communication method according to some aspects of the present disclosure.

FIG. 10 is a flow diagram of a communication method according to some aspects of the present disclosure.

DETAILED DESCRIPTION

The detailed description set forth below, in connection with the appended drawings, is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

This disclosure relates generally to wireless communications systems, also referred to as wireless communications networks. In various instances, the techniques and apparatus may be used for wireless communication networks such as code division multiple access (CDMA) networks, time divi-

sion multiple access (TDMA) networks, frequency division multiple access (FDMA) networks, orthogonal FDMA (OFDMA) networks, single-carrier FDMA (SC-FDMA) networks, LTE networks, GSM networks, 5th Generation (5G) or new radio (NR) networks, as well as other communications networks. As described herein, the terms “networks” and “systems” may be used interchangeably.

An OFDMA network may implement a radio technology such as evolved UTRA (E-UTRA), Institute of Electrical and Electronic Engineers (IEEE) 802.11, IEEE 802.16, IEEE 802.20, flash-OFDM and the like. UTRA, E-UTRA, and Global System for Mobile Communications (GSM) are part of universal mobile telecommunication system (UMTS). In particular, long term evolution (LTE) is a release of UMTS that uses E-UTRA. UTRA, E-UTRA, GSM, UMTS and LTE are described in documents provided from an organization named “3rd Generation Partnership Project” (3GPP), and cdma2000 is described in documents from an organization named “3rd Generation Partnership Project 2” (3GPP2). These various radio technologies and standards are known or are being developed. For example, the 3rd Generation Partnership Project (3GPP) is a collaboration between groups of telecommunications associations that aims to define a globally applicable third generation (3G) mobile phone specification. 3GPP long term evolution (LTE) is a 3GPP project which was aimed at improving the universal mobile telecommunications system (UMTS) mobile phone standard. The 3GPP may define specifications for the next generation of mobile networks, mobile systems, and mobile devices. The present disclosure is concerned with the evolution of wireless technologies from LTE, 4G, 5G, NR, and beyond with shared access to wireless spectrum between networks using a collection of new and different radio access technologies or radio air interfaces.

In particular, 5G networks contemplate diverse deployments, diverse spectrum, and diverse services and devices that may be implemented using an OFDM-based unified, air interface. In order to achieve these goals, further enhancements to LTE and LTE-A are considered in addition to development of the new radio technology for 5G NR networks. The 5G NR will be capable of scaling to provide coverage (1) to a massive Internet of things (IoTs) with an ultra-high density (e.g., ~1M nodes/km²), ultra-low complexity (e.g., ~10 s of bits/sec), ultra-low energy (e.g., ~10+ years of battery life), and deep coverage with the capability to reach challenging locations; (2) including mission-critical control with strong security to safeguard sensitive personal, financial, or classified information, ultra-high reliability (e.g., ~99.9999% reliability), ultra-low latency (e.g., ~1 ms), and users with wide ranges of mobility or lack thereof; and (3) with enhanced mobile broadband including extreme high capacity (e.g., ~10 Tbps/km²), extreme data rates (e.g., multi-Gbps rate, 100+ Mbps user experienced rates), and deep awareness with advanced discovery and optimizations.

The 5G NR may be implemented to use optimized OFDM-based waveforms with scalable numerology and transmission time interval (TTI); having a common, flexible framework to efficiently multiplex services and features with a dynamic, low-latency time division duplex (TDD)/frequency division duplex (FDD) design; and with advanced wireless technologies, such as massive multiple input, multiple output (MIMO), robust millimeter wave (mmWave) transmissions, advanced channel coding, and device-centric mobility. Scalability of the numerology in 5G NR, with scaling of subcarrier spacing, may efficiently address operating diverse services across diverse spectrum and diverse deployments. For example, in various outdoor and macro

coverage deployments of less than 3 GHz FDD/TDD implementations, subcarrier spacing may occur with 15 kHz, for example over 5, 10, 20 MHz, and the like bandwidth (BW). For other various outdoor and small cell coverage deployments of TDD greater than 3 GHz, subcarrier spacing may occur with 30 kHz over 80/100 MHz BW. For other various indoor wideband implementations, using a TDD over the unlicensed portion of the 5 GHz band, the subcarrier spacing may occur with 60 kHz over a 160 MHz BW. Finally, for various deployments transmitting with mmWave components at a TDD of 28 GHz, subcarrier spacing may occur with 120 kHz over a 500 MHz BW.

The scalable numerology of the 5G NR facilitates scalable TTI for diverse latency and quality of service (QoS) requirements. For example, shorter TTI may be used for low latency and high reliability, while longer TTI may be used for higher spectral efficiency. The efficient multiplexing of long and short TTIs to allow transmissions to start on symbol boundaries. 5G NR also contemplates a self-contained integrated subframe design with uplink/downlink scheduling information, data, and acknowledgement in the same subframe. The self-contained integrated subframe supports communications in unlicensed or contention-based shared spectrum, adaptive uplink/downlink that may be flexibly configured on a per-cell basis to dynamically switch between uplink and downlink to meet the current traffic needs.

Various other aspects and features of the disclosure are further described below. It should be apparent that the teachings herein may be embodied in a wide variety of forms and that any specific structure, function, or both being disclosed herein is merely representative and not limiting. Based on the teachings herein one of an ordinary level of skill in the art should appreciate that an aspect disclosed herein may be implemented independently of any other aspects and that two or more of these aspects may be combined in various ways. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, such an apparatus may be implemented or such a method may be practiced using other structure, functionality, or structure and functionality in addition to or other than one or more of the aspects set forth herein. For example, a method may be implemented as part of a system, device, apparatus, and/or as instructions stored on a computer readable medium for execution on a processor or computer. Furthermore, an aspect may include at least one element of a claim.

The deployment of NR over an unlicensed spectrum is referred to as NR-unlicensed (NR-U). Federal Communications Commission (FCC) and European Telecommunications Standards Institute (ETSI) are working on regulating 6 GHz as a new unlicensed band for wireless communications. The addition of 6 GHz bands allows for hundreds of megahertz (MHz) of bandwidth (BW) available for unlicensed band communications. Additionally, NR-U may also be deployed over 2.4 GHz unlicensed bands, which are currently shared by various radio access technologies (RATs), such as IEEE 802.11 wireless local area network (WLAN) or WiFi and/or license assisted access (LAA). Sidelink communications may benefit from utilizing the additional bandwidth available in an unlicensed spectrum. However, channel access in a certain unlicensed spectrum may be regulated by authorities. For instance, some unlicensed bands may impose restrictions on the power spectral density (PSD) and/or minimum occupied channel bandwidth (OCB) for transmissions in the unlicensed bands. For example, the unlicensed national information infrastructure

(UNII) radio band has a minimum OCB requirement of about at least 70 percent (%).

Some sidelink systems may operate over a 20 MHz bandwidth, e.g., for listen before talk (LBT) based channel accessing, in an unlicensed band. A BS may configure a sidelink resource pool over one or multiple 20 MHz LBT sub-bands for sidelink communications. A sidelink resource pool is typically allocated with multiple frequency subchannels within a sidelink band width part (SL-BWP) and a sidelink UE may select a sidelink resource (e.g., one or multiple subchannel) in frequency and one or multiple slots in time) from the sidelink resource pool for sidelink communication.

Deployment of communication systems, such as 5G new radio (NR) systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also may be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which may enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, may be configured for wired or wireless communication with at least one other unit.

Various aspects relate generally to wireless communication and more particularly to signaling for dynamic waveform switching. Some aspects more specifically relate to a network unit signaling a user equipment (UE) to switch between a first waveform type and a second waveform type for uplink communications. In some examples, a network unit may transmit an indicator to the UE to enable switching between the waveform types. When waveform switching is enabled, the network unit may transmit DCI to the UE

indicating which waveform type to use for uplink communications. In some examples, the size of the DCI may be the same size for the first waveform type and the second waveform type. As such, the UE may blind decode the DCI using a common DCI size for the first waveform type and the second waveform type. The DCI may further include scheduled resources for a physical uplink shared channel (PUSCH) communication associated with the UE. The UE may transmit PUSCH communications to the network unit via the scheduled resources using the indicated waveform type.

Additionally or alternatively, the UE may switch between the first waveform type and the second waveform type on a semi-static basis. In some examples, a network unit may transmit an indicator to the UE to enable switching between the waveform types. When waveform switching is enabled, the network unit may transmit non-uplink scheduling DCI and/or a MAC-CE communication to the UE indicating which waveform type to use for uplink communications. The network unit may subsequently transmit uplink scheduling DCI to the UE using a DCI size associated with the previously indicated waveform type. The DCI size associated with the first waveform type may be different from the DCI associated with the second waveform type. As such, the UE may blind decode the DCI based on the DCI size associated with the indicated waveform type. The UE may transmit PUSCH communications to the network unit via the scheduled resources using the indicated waveform type.

Particular aspects of the subject matter described in this disclosure may be implemented to realize one or more of the following potential advantages. In some examples, by implementing dynamic waveform switching according to embodiments of the present disclosure, the described techniques may be used to reduce computing resources, memory requirements, latency, and/or power consumption in the UE by blind decoding a DCI having a common size for the first and second waveform types as compared to blind decoding a first DCI associated with the first waveform type and blind decoding a second, different sized DCI associated with the second waveform type. The dynamic waveform switching according to embodiments of the present disclosure may increase network coverage and/or network capacity. For example, the UE may switch to transmitting uplink communications using a DFT-s-OFDM waveform to increase range and coverage. In some examples, the UE may switch to transmitting uplink communications using a CP-OFDM waveform to increase throughput and/or data rate.

FIG. 1 illustrates a wireless communication network 100 according to some aspects of the present disclosure. The network 100 includes a number of base stations (BSs) 105 and other network entities. A BS 105 may be a station that communicates with UEs 115 and may also be referred to as an evolved node B (eNB), a next generation eNB (gNB), an access point, and the like. Each BS 105 may provide communication coverage for a particular geographic area. In 3GPP, the term "cell" may refer to this particular geographic coverage area of a BS 105 and/or a BS subsystem serving the coverage area, depending on the context in which the term is used.

A BS 105 may provide communication coverage for a macro cell or a small cell, such as a pico cell or a femto cell, and/or other types of cell. A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs with service subscriptions with the network provider. A small cell, such as a pico cell, would generally cover a relatively smaller geographic area and may allow unrestricted access

by UEs with service subscriptions with the network provider. A small cell, such as a femto cell, would also generally cover a relatively small geographic area (e.g., a home) and, in addition to unrestricted access, may also provide restricted access by UEs having an association with the femto cell (e.g., UEs in a closed subscriber group (CSG), UEs for users in the home, and the like). A BS for a macro cell may be referred to as a macro BS. A BS for a small cell may be referred to as a small cell BS, a pico BS, a femto BS or a home BS. In the example shown in FIG. 1, the BSs 105d and 105e may be regular macro BSs, while the BSs 105a-105c may be macro BSs enabled with one of three dimension (3D), full dimension (FD), or massive MIMO. The BSs 105a-105c may take advantage of their higher dimension MIMO capabilities to exploit 3D beamforming in both elevation and azimuth beamforming to increase coverage and capacity. The BS 105f may be a small cell BS which may be a home node or portable access point. A BS 105 may support one or multiple (e.g., two, three, four, and the like) cells.

The network 100 may support synchronous or asynchronous operation. For synchronous operation, the BSs may have similar frame timing, and transmissions from different BSs may be approximately aligned in time. For asynchronous operation, the BSs may have different frame timing, and transmissions from different BSs may not be aligned in time.

The UEs 115 are dispersed throughout the wireless network 100, and each UE 115 may be stationary or mobile. A UE 115 may also be referred to as a terminal, a mobile station, a subscriber unit, a station, or the like. A UE 115 may be a cellular phone, a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a tablet computer, a laptop computer, a cordless phone, a wireless local loop (WLL) station, or the like. In one aspect, a UE 115 may be a device that includes a Universal Integrated Circuit Card (UICC). In another aspect, a UE may be a device that does not include a UICC. In some aspects, the UEs 115 that do not include UICCs may also be referred to as IoT devices or internet of everything (IoE) devices. The UEs 115a-115d are examples of mobile smart phone-type devices accessing network 100. A UE 115 may also be a machine specifically configured for connected communication, including machine type communication (MTC), enhanced MTC (eMTC), narrowband IoT (NB-IoT) and the like. The UEs 115e-115h are examples of various machines configured for communication that access the network 100. The UEs 115i-115k are examples of vehicles equipped with wireless communication devices configured for communication that access the network 100. A UE 115 may be able to communicate with any type of the BSs, whether macro BS, small cell, or the like. In FIG. 1, a lightning bolt (e.g., communication links) indicates wireless transmissions between a UE 115 and a serving BS 105, which is a BS designated to serve the UE 115 on the downlink (DL) and/or uplink (UL), desired transmission between BSs 105, backhaul transmissions between BSs, or sidelink transmissions between UEs 115.

In operation, the BSs 105a-105c may serve the UEs 115a and 115b using 3D beamforming and coordinated spatial techniques, such as coordinated multipoint (CoMP) or multi-connectivity. The macro BS 105d may perform backhaul communications with the BSs 105a-105c, as well as small cell, the BS 105f. The macro BS 105d may also transmits multicast services which are subscribed to and received by the UEs 115c and 115d. Such multicast services may include mobile television or stream video, or may

include other services for providing community information, such as weather emergencies or alerts, such as Amber alerts or gray alerts.

The BSs 105 may also communicate with a core network. The core network may provide user authentication, access authorization, tracking, Internet Protocol (IP) connectivity, and other access, routing, or mobility functions. At least some of the BSs 105 (e.g., which may be an example of an evolved NodeB (eNB) or an access node controller (ANC)) may interface with the core network 130 through backhaul links (e.g., S1, S2, etc.) and may perform radio configuration and scheduling for communication with the UEs 115. In various examples, the BSs 105 may communicate, either directly or indirectly (e.g., through core network), with each other over backhaul links (e.g., X1, X2, etc.), which may be wired or wireless communication links.

The network 100 may also support mission critical communications with ultra-reliable and redundant links for mission critical devices, such as the UE 115e, which may be a vehicle (e.g., a car, a truck, a bus, an autonomous vehicle, an aircraft, a boat, etc.). Redundant communication links with the UE 115e may include links from the macro BSs 105d and 105e, as well as links from the small cell BS 105f. Other machine type devices, such as the UE 115f (e.g., a thermometer), the UE 115g (e.g., smart meter), and UE 115h (e.g., wearable device) may communicate through the network 100 either directly with BSs, such as the small cell BS 105f, and the macro BS 105e, or in multi-hop configurations by communicating with another user device which relays its information to the network, such as the UE 115f communicating temperature measurement information to the smart meter, the UE 115g, which is then reported to the network through the small cell BS 105f. In some aspects, the UE 115h may harvest energy from an ambient environment associated with the UE 115h. The network 100 may also provide additional network efficiency through dynamic, low-latency TDD/FDD communications, such as vehicle-to-vehicle (V2V), vehicle-to-everything (V2X), cellular-vehicle-to-everything (C-V2X) communications between a UE 115i, 115j, or 115k and other UEs 115, and/or vehicle-to-infrastructure (V2I) communications between a UE 115i, 115j, or 115k and a BS 105.

In some implementations, the network 100 utilizes OFDM-based waveforms for communications. An OFDM-based system may partition the system BW into multiple (K) orthogonal subcarriers, which are also commonly referred to as subcarriers, tones, bins, or the like. Each subcarrier may be modulated with data. In some instances, the subcarrier spacing between adjacent subcarriers may be fixed, and the total number of subcarriers (K) may be dependent on the system BW. The system BW may also be partitioned into subbands. In other instances, the subcarrier spacing and/or the duration of TTIs may be scalable.

In some instances, the BSs 105 may assign or schedule transmission resources (e.g., in the form of time-frequency resource blocks (RB)) for downlink (DL) and uplink (UL) transmissions in the network 100. DL refers to the transmission direction from a BS 105 to a UE 115, whereas UL refers to the transmission direction from a UE 115 to a BS 105. The communication may be in the form of radio frames. A radio frame may be divided into a plurality of subframes, for example, about 10. Each subframe may be divided into slots, for example, about 2. Each slot may be further divided into mini-slots. In a FDD mode, simultaneous UL and DL transmissions may occur in different frequency bands. For example, each subframe includes a UL subframe in a UL frequency band and a DL subframe in a DL frequency band.

In a TDD mode, UL and DL transmissions occur at different time periods using the same frequency band. For example, a subset of the subframes (e.g., DL subframes) in a radio frame may be used for DL transmissions and another subset of the subframes (e.g., UL subframes) in the radio frame may be used for UL transmissions.

The DL subframes and the UL subframes may be further divided into several regions. For example, each DL or UL subframe may have pre-defined regions for transmissions of reference signals, control information, and data. Reference signals are predetermined signals that facilitate the communications between the BSs 105 and the UEs 115. For example, a reference signal may have a particular pilot pattern or structure, where pilot tones may span across an operational BW or frequency band, each positioned at a pre-defined time and a pre-defined frequency. For example, a BS 105 may transmit cell specific reference signals (CRSs) and/or channel state information—reference signals (CSI-RSs) to enable a UE 115 to estimate a DL channel. Similarly, a UE 115 may transmit sounding reference signals (SRSs) to enable a BS 105 to estimate a UL channel. Control information may include resource assignments and protocol controls. Data may include protocol data and/or operational data. In some instances, the BSs 105 and the UEs 115 may communicate using self-contained subframes. A self-contained subframe may include a portion for DL communication and a portion for UL communication. A self-contained subframe may be DL-centric or UL-centric. A DL-centric subframe may include a longer duration for DL communication than for UL communication. A UL-centric subframe may include a longer duration for UL communication than for UL communication.

In some instances, the network 100 may be an NR network deployed over a licensed spectrum. The BSs 105 may transmit synchronization signals (e.g., including a primary synchronization signal (PSS) and a secondary synchronization signal (SSS)) in the network 100 to facilitate synchronization. The BSs 105 may broadcast system information associated with the network 100 (e.g., including a master information block (MIB), remaining minimum system information (RMSI), and other system information (OSI)) to facilitate initial network access. In some instances, the BSs 105 may broadcast the PSS, the SSS, and/or the MIB in the form of synchronization signal blocks (SSBs) over a physical broadcast channel (PBCH) and may broadcast the RMSI and/or the OSI over a physical downlink shared channel (PDSCH).

In some instances, a UE 115 attempting to access the network 100 may perform an initial cell search by detecting a PSS from a BS 105. The PSS may enable synchronization of period timing and may indicate a physical layer identity value. The UE 115 may then receive an SSS. The SSS may enable radio frame synchronization, and may provide a cell identity value, which may be combined with the physical layer identity value to identify the cell. The SSS may also enable detection of a duplexing mode and a cyclic prefix length. The PSS and the SSS may be located in a central portion of a carrier or any suitable frequencies within the carrier.

After receiving the PSS and SSS, the UE 115 may receive a MIB. The MIB may include system information for initial network access and scheduling information for RMSI and/or OSI. After decoding the MIB, the UE 115 may receive RMSI and/or OSI. The RMSI and/or OSI may include radio resource control (RRC) information related to random access channel (RACH) procedures, paging, control resource set (CORESET) for physical downlink control

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channel (PDCCH) monitoring, physical uplink control channel (PUCCH), physical uplink shared channel (PUSCH), power control, SRS, and cell barring.

After obtaining the MIB, the RMSI and/or the OSI, the UE 115 may perform a random access procedure to establish a connection with the BS 105. For the random access procedure, the UE 115 may transmit a random access preamble and the BS 105 may respond with a random access response. Upon receiving the random access response, the UE 115 may transmit a connection request to the BS 105 and the BS 105 may respond with a connection response (e.g., contention resolution message).

After establishing a connection, the UE 115 and the BS 105 may enter a normal operation stage, where operational data may be exchanged. For example, the BS 105 may schedule the UE 115 for UL and/or DL communications. The BS 105 may transmit UL and/or DL scheduling grants to the UE 115 via a PDCCH. The BS 105 may transmit a DL communication signal to the UE 115 via a PDSCH according to a DL scheduling grant. The UE 115 may transmit a UL communication signal to the BS 105 via a PUSCH and/or PUCCH according to a UL scheduling grant.

The network 100 may be designed to enable a wide range of use cases. While in some examples a network 100 may utilize monolithic base stations, there are a number of other architectures which may be used to perform aspects of the present disclosure. For example, a BS 105 may be separated into a remote radio head (RRH) and baseband unit (BBU). BBUs may be centralized into a BBU pool and connected to RRHs through low-latency and high-bandwidth transport links, such as optical transport links. BBU pools may be cloud-based resources. In some aspects, baseband processing is performed on virtualized servers running in data centers rather than being co-located with a BS 105. In another example, based station functionality may be split between a remote unit (RU), distributed unit (DU), and a central unit (CU). An RU generally performs low physical layer functions while a DU performs higher layer functions, which may include higher physical layer functions. A CU performs the higher RAN functions, such as radio resource control (RRC).

For simplicity of discussion, the present disclosure refers to methods of the present disclosure being performed by base stations, or more generally network entities, while the functionality may be performed by a variety of architectures other than a monolithic base station. In addition to disaggregated base stations, aspects of the present disclosure may also be performed by a centralized unit (CU), a distributed unit (DU), a radio unit (RU), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), a Non-Real Time (Non-RT) RIC, integrated access and backhaul (IAB) node, a relay node, a sidelink node, etc.

In some aspects, the BS 105 may perform a first listen-before-talk (LBT) procedure in a shared frequency band, wherein a category of the first LBT procedure is based on a first channel access priority class (CAPC) index value associated with a plurality of transport blocks (TBs). The BS 105 may transmit, to the UE 115 based on the first LBT procedure being successful, one or more TBs of the plurality of TBs. The BS 105 may perform a second LBT procedure, wherein at least one parameter of the second LBT procedure is based on a survival time associated with the one or more TBs.

In some aspects, the UE 115 may transmit, to the network unit 105, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array

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at the UE 115 for uplink transmissions. The UE 115 may receive, from the network unit 105, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array. The UE 115 may transmit, to the network unit 105, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

In some aspects, the UE 115 may transmit, to a network unit 105, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE 115 for uplink transmissions. The UE 115 may receive, from the network unit 105, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array. The UE 115 may receive, from the network unit 105, signals used to determine the calibration coefficients associated with a second set of phase and gain values. The UE 115 may transmit, to the network unit 105, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values.

FIG. 2 shows a diagram illustrating an example disaggregated base station 200 architecture. The disaggregated base station 200 architecture may include one or more central units (CUs) 210 that may communicate directly with a core network 220 via a backhaul link, or indirectly with the core network 220 through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) 225 via an E2 link, or a Non-Real Time (Non-RT) RIC 215 associated with a Service Management and Orchestration (SMO) Framework 205, or both). A CU 210 may communicate with one or more distributed units (DUs) 230 via respective midhaul links, such as an F1 interface. The DUs 230 may communicate with one or more radio units (RUs) 240 via respective fronthaul links. The RUs 240 may communicate with respective UEs 115 via one or more radio frequency (RF) access links. In some implementations, the UE 115 may be simultaneously served by multiple RUs 240.

Each of the units, i.e., the CUs 210, the DUs 230, the RUs 240, as well as the Near-RT RICs 225, the Non-RT RICs 215 and the SMO Framework 205, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, may be configured to communicate with one or more of the other units via the transmission medium. For example, the units may include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units may include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a radio frequency (RF) transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

In some aspects, the CU 210 may host one or more higher layer control functions. Such control functions may include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP),

or the like. Each control function may be implemented with an interface configured to communicate signals with other control functions hosted by the CU 210. The CU 210 may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU 210 may be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit may communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU 210 may be implemented to communicate with the DU 230, as necessary, for network control and signaling.

The DU 230 may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs 240. In some aspects, the DU 230 may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3rd Generation Partnership Project (3GPP). In some aspects, the DU 230 may further host one or more low PHY layers. Each layer (or module) may be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU 230, or with the control functions hosted by the CU 210.

Lower-layer functionality may be implemented by one or more RUs 240. In some deployments, an RU 240, controlled by a DU 230, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) 240 may be implemented to handle over the air (OTA) communication with one or more UEs 115. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) 240 may be controlled by the corresponding DU 230. In some scenarios, this configuration may enable the DU(s) 230 and the CU 210 to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

The SMO Framework 205 may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework 205 may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework 205 may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) 290) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements may include CUs 210, DUs 230, RUs 240 and Near-RT RICs 225. In some implementations, the SMO Framework 205 may communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) 211, via an O1 interface. Additionally, in some implementations, the SMO Framework 205 may communicate directly with one or more RUs 240 via an O1 interface.

The SMO Framework 205 also may include a Non-RT RIC 215 configured to support functionality of the SMO Framework 205.

The Non-RT RIC 215 may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC 225. The Non-RT RIC 215 may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC 225. The Near-RT RIC 225 may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs 210, one or more DUs 230, or both, as well as an O-eNB, with the Near-RT RIC 225.

In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC 225, the Non-RT RIC 215 may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC 225 and may be received at the SMO Framework 205 or the Non-RT RIC 215 from non-network data sources or from network functions. In some examples, the Non-RT RIC 215 or the Near-RT RIC 225 may be configured to tune RAN behavior or performance. For example, the Non-RT RIC 215 may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework 205 (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

In some aspects, the UE 115 may transmit, to the RU 240, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE 115 for uplink transmissions. The UE 115 may receive, from the RU 240, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array. The UE 115 may transmit, to the RU 240, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

In some aspects, the UE 115 may transmit, to a RU 240, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE 115 for uplink transmissions. The UE 115 may receive, from the network unit 105, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array. The UE 115 may receive, from the network unit 105, signals used to determine the calibration coefficients associated with a second set of phase and gain values. The UE 115 may transmit, to the RU 240, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values.

FIG. 3 illustrates joint phase/gain calibration values in wireless communications according to some aspects of the present disclosure. In some aspects, a user equipment (UE) (e.g., the UE 115, the UE 700), may transmit a plurality of reference signals to a network unit (e.g., the network unit 800, the BS 105, the RU 240, the DU 230, and/or the CU

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210). In this regard the UE may transmit a plurality of sounding reference signals (SRSs), demodulation reference signals (DMRSs), phase tracking reference signals (PTRSs), or other suitable reference signals to the network unit. In some aspects, the plurality of reference signals is associated with a first set of phase values and gain values to be used with a first antenna array. The first antenna array may be an antenna array of one or more antenna arrays used by the UE for uplink transmissions to the network unit. In this regard, the uplink transmissions may include PUCCH transmissions, PUSCH transmissions, PRACH transmissions, and/or other suitable uplink transmissions.

In some aspects, the UE may include one or more antenna arrays at multiple positions in the UE. For example, the antenna arrays may be positioned along different edges (e.g., sides) of the UE for spatial diversity. The antenna arrays may be positioned parallel and/or orthogonal to one another. Each antenna array may include multiple antenna elements. For example, each antenna array may include a single row of antenna elements and/or multiple rows of antenna elements.

In some aspects, the UE may receive signals from the network unit. The signals received by the UE from the network unit may be used to determine calibration coefficients associated with the first set of phase values and gain values for the first antenna array. The calibration coefficients may be used by the UE to increase a signal quality of a transmitted beam for uplink transmissions based on signals received on the downlink. For example, signal processing over an array of antenna elements during beamforming may include adjusting a phase of signals corresponding to each of the antenna elements in the array. In some aspects, gain adjustments may be used for improving directionality of beams from the antenna elements, such as to control (e.g., reduce) side lobes and control (e.g., increase) the main lobe relatively. Combining gain adjustments with phase adjustments in beamforming may be particularly advantageous in millimeter wave communications to coherently combine energy and overcome high path, propagation, and blockage losses at the higher frequencies of millimeter wave communications. Upon performing gain and phase calibration for beamforming, the antenna elements may operate with more accurate gain and phase settings for a more accurate beam pattern.

In some aspects, the UE may transmit a communication signal (e.g., UCI, a PUCCH communication, a PUSCH communication, or other suitable uplink communication) to the network unit based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values. In some aspects, the calibration coefficients associated with the second set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values. In this regard, the calibration coefficients associated with the second set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values using a linear interpolation method, a sample and hold interpolation method, a spline interpolation method, or other suitable interpolation method.

In some aspects, the calibration coefficients associated with the second set of phase values and gain values may be interpolated from adjacent calibration coefficients associated with the first set of phase values and gain values. In some aspects, the first and second sets of calibration coefficients may be stored in memory (e.g., memory 704, 804) in the

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form of a two dimensional table. Referring to FIG. 3, each antenna element 306a to 306n of the antenna may be represented by a different two dimensional table 300. Each two dimensional table corresponding to each of the antenna elements 306a to 306n may be configured with phase/gain calibration coefficients. The number of columns in the table 300 may be based on the number of phase settings. For example, if the UE includes an X-bit phase shifter, the number of columns may be 2^X . The number of rows in the table 300 may be based on the number of gain settings. For example, if the UE includes a Y-bit gain controller, the number of rows may be 2^Y . In a non-limiting example as shown in FIG. 3, the UE may include a 3 bit phase shifter and a 3 bit gain controller such that the table of calibration coefficients includes 8 rows and 8 columns. Each calibration coefficient in the table 300 may correspond to a row index labeled from G0 to G7 and a column index labeled from P0 to P7. In some aspects, the calibration coefficients associated with the first set of phase values and gain values generated by the over-the-air calibration process may be represented in a first subset of calibration coefficients in the table 300. The calibration coefficients associated with the second set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values and stored in a second subset of row/column entries. For example, as shown in FIG. 3, the patterned entries in the table 300 may include calibrated values 302. The unpatterned entries in the table 300 may include interpolated values 304.

In the example of FIG. 3, the interpolated values 304 may be interpolated from one or more calibrated values 302. In a non-limiting example, the interpolated value may be based on one or more of the calibrated values in proximity to interpolated values. In some instances, the calibrated value(s) used to determine the interpolated value may include one or more calibrated values directly adjacent to the interpolated value, one or more calibrated values nearby but spaced from the interpolated value, one or more contiguous calibrated values directly adjacent and/or spaced from the interpolated values, one or more non-contiguous calibrated values, one or more calibrated values surrounding (directly or indirectly) the interpolated value, and/or any other calibrated values. As an example, the interpolated value at G1/P1 may be interpolated from one or more of directly adjacent calibrated values at G1/P0, G2/P1, G1/P2, and/or G0/P1. As another example, the interpolated value at G2/P2 may be interpolated from one or more of calibrated values at G1/P0, G3/P0, G3/P4, and/or G1/P4 that are nearby but spaced from G2/P2. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values surrounding G4/P4, such as G4/P1, G2/P1, G2/P3, G2/P5, G3/P6, G5/P6, G6/P5, G6/P3, etc. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values contiguous to G4/P4, such as G4/P3, G3/P4, G4/P5, and G5/P4. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values non-contiguous to G4/P4, such as G4/P1, G2/P3, G4/P7, and G7/P4. In some aspects, the interpolated values 304 may be interpolated from any number of calibrated values 302 at any position within the table 300a.

In some aspects, the UE may determine the number of calibrated values 302 and the number of interpolated values 304 in the table 300. The number of calibrated values 302 and/or the number of interpolated values 304 may be based on a battery power level of the UE, a temperature of the UE, a frequency range associated with the communication sig-

nal, or other suitable parameter. In some aspects, the UE may transmit an indicator to the network unit indicating the number of calibrated values **302** and/or the number of interpolated values **304**. In this regard, the UE may transmit the indicator to the network unit via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. Additionally or alternatively, the network unit may transmit an indicator to the UE indicating the number of calibrated values **302** and/or the number of interpolated values **304**. In this regard, the network unit may transmit the indicator to the UE via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication.

In some aspects, the antenna may include n antenna elements **306**. FIG. 3 shows the same configuration for each antenna element **306a** to **306n** in which the tables **300a** to **300n** have the same number of calibrated values **302** and interpolated values **304**. However, the present disclosure is not so limited and each antenna element **306a** to **306n** may have a different number of calibrated values **302** and interpolated values **304**.

FIG. 4 illustrates joint phase/gain calibration values in wireless communications according to some aspects of the present disclosure. In some aspects, a user equipment (UE) (e.g., the UE **115**, the UE **700**), may transmit a plurality of reference signals to a network unit (e.g., the network unit **800**, the BS **105**, the RU **240**, the DU **230**, and/or the CU **210**). In this regard, the UE may transmit a plurality of sounding reference signals (SRSs), demodulation reference signals (DMRSs), phase tracking reference signals (PTRSs), or other suitable reference signals to the network unit. In some aspects, the plurality of reference signals is associated with a first set of phase values and gain values to be used with a first antenna array. The first antenna array may be an antenna array of one or more antenna arrays used by the UE for uplink transmissions to the network unit.

In some aspects, the UE may receive signals from the network unit. The signals received by the UE from the network unit may be used to determine calibration coefficients associated with the first set of phase values and gain values for the first antenna array. The calibration coefficients may be used by the UE to increase a signal quality of a transmitted beam for uplink transmissions. For example, signal processing over an array of antenna elements during beamforming may include adjusting a phase of signals corresponding to each of the antenna elements in the array. In some aspects, gain adjustments may be used for improving directionality of beams from the antenna elements, such as to control (e.g., reduce) side lobes and control (e.g., increase) the main lobe. Combining gain adjustments with phase adjustments in beamforming may be particularly advantageous in millimeter wave communications to coherently combine energy and overcome high path, propagation, and blockage losses at the higher frequencies of millimeter wave communications. Upon performing gain and phase calibration for beamforming, the antenna elements may operate with more accurate gain and phase settings for a more accurate beam pattern.

In some aspects, the UE may receive calibration coefficients associated with a second set of phase and gain values. In this regard, the UE may receive the second set of phase and gain values from the network unit via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication. In some aspects, the network unit may receive the second set of phase and gain values from one or more additional UEs. For example, one or more additional UEs may transmit a set of calibration coefficients to the

network unit. The network unit may transmit the set or a subset of the calibration coefficients to the UE. The network unit may use a federated (e.g., crowdsourcing) method of receiving the calibration coefficients from the additional UE(s). The calibration coefficients associated with the additional UEs may be the same or similar (e.g., correlated) to the calibration coefficients associated with the UE. The additional UE(s) may be of a same type and/or class as the UE. For example, the additional UEs may have a same model type as the UE. In some aspects, the additional UEs may have the same components (e.g., amplifiers, phase shifters, mixers, etc.) as the UE.

In some aspects, the UE may transmit a communication signal (e.g., UCI, a PUCCH communication, a PUSCH communication, or other suitable uplink communication) to the network unit based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values. In some aspects, the calibration coefficients associated with the second set of phase values and gain values may be the calibration coefficients associated with the additional UE(s) that was received from the base station. Additionally or alternatively, the UE may transmit the uplink communication using the first set of calibration coefficients, the second set of calibration coefficients, and a third set of calibration coefficients that is interpolated from the first set of calibration coefficients. In this regard, the calibration coefficients associated with the third set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values using a linear interpolation method, a sample and hold interpolation method, a spline interpolation method, or other suitable interpolation method.

In some aspects, the calibration coefficients associated with the third set of phase values and gain values may be interpolated from adjacent calibration coefficients associated with the first set of phase values and gain values. In some aspects, the first, second, and third sets of calibration coefficients may be stored in memory (e.g., memory **704**, **804**) in the form of a two dimensional table **400**. Referring to FIG. 4, each antenna element **306a** to **306n** of the antenna may be represented by a different two dimensional table **400a** to **400n**. Each two dimensional table **400** corresponding to each of the antenna elements **306a** to **306n** may be configured with phase/gain calibration coefficients. The number of columns in the table **400** may be based on the number of phase settings. For example, if the UE includes an X -bit phase shifter the number of columns may be 2^X . The number of rows in the table **400** may be based on the number of gain settings. For example, if the UE includes a Y -bit gain controller the number of rows may be 2^Y . In a non-limiting example as shown in FIG. 4, the UE may include a 3 bit phase shifter and a 3 bit gain controller such that the table **400** of calibration coefficients includes 8 rows and 8 columns. Each calibration coefficient in the table **400** may correspond to a row index labeled from G0 to G7 and a column index labeled from P0 to P7. In some aspects, the calibration coefficients associated with the first set of phase values and gain values generated by the over the air calibration process may be represented in a first subset of calibration coefficients in the table **400**. The calibration coefficients associated with the second set of phase and gain values received from the network unit may be represented as a second subset. The calibration coefficients associated with the third set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set and/or the second set and stored in a third subset

of row/column entries. For example, as shown in FIG. 4, the cross hatched patterned entries in the table 400 may include calibrated values 302, the unpatterned entries in the table 400 may include interpolated values 304 and the diagonal patterned entries in the table 400 may include the federated values 405.

In the example of FIG. 4, the interpolated values 304 may be interpolated from one or more calibrated values 302 and/or one or more federated values 405. In a non-limiting example, the interpolated values 304 may be based on one or more of the calibrated values 302 and/or one or more federated values 405 in proximity to interpolated values 304. In some instances, the calibrated value(s) 302 used to determine the interpolated value may include one or more calibrated values and/or one or more federated values directly adjacent to the interpolated value, one or more calibrated values and/or one or more federated values nearby but spaced from the interpolated value, one or more contiguous calibrated values and/or one or more contiguous federated values directly adjacent and/or spaced from the interpolated values, one or more non-contiguous calibrated values and/or one or more non-contiguous federated values, one or more calibrated values and/or one or more federated values surrounding (directly or indirectly) the interpolated value, and/or any other calibrated values and/or any other federated values. As an example, the interpolated value at G1/P1 may be interpolated from one or more of directly adjacent calibrated values and/or one or more directly adjacent federated values at G1/P0, G2/P1, G1/P2, and/or G0/P1. As another example, the interpolated value at G2/P2 may be interpolated from one or more of calibrated values and/or one or more federated values at G1/P0, G3/P0, G3/P4, and/or G1/P4 that are nearby but spaced from G2/P2. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values and/or one or more federated values surrounding G4/P4, such as G4/P1, G2/P1, G2/P3, G2/P5, G3/P6, G5/P6, G6/P5, G6/P3, etc. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values and/or one or more federated values contiguous to G4/P4, such as G4/P3, G3/P4, G4/P5, and G5/P4. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values and/or one or more federated values non-contiguous to G4/P4, such as G4/P1, G2/P3, G4/P7, and G7/P4. In some aspects, the interpolated values 304 may be interpolated from any number of calibrated values 302 and/or any number of federated values at any position within the table 400.

In some aspects, the UE may determine the number of calibrated values 302, the number of interpolated values 304, and/or the number of federated values 405 in the table 400. The number of calibrated values 302, the number of interpolated values 304, and/or the number of federated values 405 may be based on a battery power level of the UE, a temperature of the UE, a frequency range associated with the communication signal, or other suitable parameter. In some aspects, the UE may transmit an indicator to the network unit indicating the number of calibrated values 302, the number of interpolated values 304, and/or the number of federated values 405. In this regard, the UE may transmit the indicator to the network unit via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. Additionally or alternatively, the network unit may transmit an indicator to the UE indicating the number of calibrated values 302, the number of interpolated values 304, and/or the number of federated values 405. In this regard, the network unit may transmit the indicator to

the UE via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication.

In some aspects, the antenna may include n antenna elements. FIG. 4 shows the same configuration for each antenna element 306a to 306n in which each table has the same number of calibrated values 302 and interpolated values 304. However, the present disclosure is not so limited and each antenna element 306a to 306n may have a different number of calibrated values 302, interpolated values 304, and federated values 405.

FIG. 5 is a flow diagram of a communication method 500 according to some aspects of the present disclosure. Aspects of the method 500 may be executed by a computing device (e.g., a processor, processing circuit, and/or other suitable component) of a wireless communication device or other suitable means for performing the actions. For example, a wireless communication device, such as the UE 115 or the UE 700 may utilize one or more components, such as the processor 702, the memory 704, the joint phase/gain calibration module 708, the transceiver 710, the modem 712, and the one or more antennas 716, to execute aspects of method 500. The method 500 may employ similar mechanisms as in the networks 100 and 200 and the aspects and actions described with respect to FIGS. 3-4. As illustrated, the method 500 includes a number of enumerated actions, but the method 500 may include additional actions before, after, and in between the enumerated actions. In some aspects, one or more of the enumerated actions may be omitted or performed in a different order.

At action 502, the UE 115 may transmit an antenna calibration request to the network unit 105. In this regard, the UE 115 may transmit the calibration request to the network unit 105 via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. In some aspects the phase/gain calibration of the antenna(s) may be affected by certain events associated with the UE 115. In this regard, the UE 115 may transmit the request for the antenna array calibration based on an event associated with the UE 115. In some aspects, the event may include a temperature of the UE 115 satisfying a temperature threshold. For example, the temperature sensor of the UE 115 may indicate that the temperature of the UE 115 is greater than, equal to, or less than a temperature threshold. Since the UE 115 temperature may affect the antenna(s) calibration, the UE 115 may transmit the antenna array calibration request based on the temperature of the UE 115 satisfying the temperature threshold.

In some aspects, the event may include a frequency range associated with the communication signal satisfying a frequency threshold. For example, when the UE 115 is configured to switch from a first frequency range (e.g., FR1, FR2, FR3, FR4, FR5) to a second (different) frequency range (e.g., FR1, FR2, FR3, FR4, FR5), the UE 115 may transmit the antenna array calibration request based on the difference in the first frequency range and the second frequency range satisfying a frequency range threshold.

In some aspects, the UE 115 may transmit the request for the antenna array calibration based on a mobility of the UE 115. A higher mobility rate of the UE 115 may require more frequent calibration of the antenna elements as compared to a lower mobility rate.

In some aspects, the UE 115 may transmit the request for the antenna array calibration based on a battery power level of the UE 115. The UE 115 may transmit the request for antenna calibration less frequently when the battery power level is low as compared to when the battery power level is high. Since the antenna calibration process may consume

battery power in the UE 115, the UE 115 may transmit the request for antenna calibration less frequently in order to conserve battery power.

At action 504, the network unit 105 may transmit an antenna calibration acknowledgement to the UE 115. In this regard, the network unit 105 may transmit the calibration acknowledgment to the UE 115 via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication. The network unit 105 may respond to the request at action 502 by transmitting an antenna array calibration acknowledgment to the UE 115.

At action 506, the UE 115 may transmit a plurality of reference signals to the network unit 105. In this regard the UE 115 may transmit a plurality of sounding reference signals (SRSs), demodulation reference signals (DMRSs), phase tracking reference signals (PTRSs), or other suitable reference signals to the network unit 105. In some aspects, the plurality of reference signals is associated with a first set of phase values and gain values to be used with a first antenna array.

At action 508, the network unit 105 may transmit signals to the UE 115. The signals received by the UE 115 from the network unit 105 may be used to determine calibration coefficients associated with the first set of phase values and gain values for the first antenna array. The calibration coefficients may be used by the UE 115 to increase a signal quality of a transmitted beam for uplink transmissions. Combining gain adjustments with phase adjustments in beamforming may be particularly advantageous in millimeter wave communications to coherently combine energy and overcome high path, propagation, and blockage losses at the higher frequencies of millimeter wave communications. Upon performing gain and phase calibration for beamforming, the antenna elements may operate with more accurate gain and phase settings for a more accurate beam pattern.

At action 510, the UE 115 may interpolate a second set of phase/gain calibration coefficients from the first set of phase/gain calibration coefficients. The UE 115 may interpolate the second set of phase/gain calibration coefficients from the first set of phase/gain calibration coefficients using the methods described with reference to FIG. 3. In this regard, the calibration coefficients associated with the second set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values using a linear interpolation method, a sample and hold interpolation method, a spline interpolation method, or other suitable interpolation method.

At action 512, the UE 115 may transmit a communication signal (e.g., UCI, a PUCCH communication, a PUSCH communication, or other suitable uplink communication) to the network unit 105 based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with the second set of phase and gain values.

FIG. 6 is a flow diagram of a communication method 600 according to some aspects of the present disclosure. Aspects of the method 600 may be executed by a computing device (e.g., a processor, processing circuit, and/or other suitable component) of a wireless communication device or other suitable means for performing the actions. For example, a wireless communication device, such as the UE 115 or the UE 700 may utilize one or more components, such as the processor 702, the memory 704, the joint phase/gain calibration module 708, the transceiver 710, the modem 712, and the one or more antennas 716, to execute aspects of method 600. The method 600 may employ similar mechanisms as in the networks 100 and 200 and the aspects and

actions described with respect to FIGS. 3-4. As illustrated, the method 600 includes a number of enumerated actions, but the method 600 may include additional actions before, after, and in between the enumerated actions. In some aspects, one or more of the enumerated actions may be omitted or performed in a different order.

At action 602, the UE 115b may transmit an antenna calibration request to the network unit 105. In this regard, the UE 115b may transmit the calibration request to the network unit 105 via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. In some aspects the phase/gain calibration of the antenna(s) may be affected by certain events associated with the UE 115b. In this regard, the UE 115b may transmit the request for the antenna array calibration based on an event associated with the UE 115b. In some aspects, the event may include a temperature of the UE 115b satisfying a temperature threshold. For example, the temperature sensor of the UE 115b may indicate that the temperature of the UE 115b is greater than, equal to, or less than a temperature threshold. Since the UE 115b temperature may affect the antenna(s) calibration, the UE 115b may transmit the antenna array calibration request based on the temperature of the UE 115b satisfying the temperature threshold.

In some aspects, the event may include a frequency range associated with the communication signal satisfying a frequency threshold. For example, when the UE 115b is configured to switch from a first frequency range (e.g., FR1, FR2, FR3, FR4, FR5) to a second (different) frequency range (e.g., FR1, FR2, FR3, FR4, FR5), the UE 115b may transmit the antenna array calibration request based on the difference in the first frequency range and the second frequency range satisfying a frequency range threshold.

In some aspects, the UE 115b may transmit the request for the antenna array calibration based on a mobility of the UE 115b. A higher mobility rate of the UE 115b may require more frequent calibration of the antenna elements as compared to a lower mobility rate.

In some aspects, the UE 115b may transmit the request for the antenna array calibration based on a battery power level of the UE 115b. The UE 115b may transmit the request for antenna calibration less frequently when the battery power level is low as compared to when the battery power level is high. Since the antenna calibration process may consume battery power in the UE 115b, the UE 115b may transmit the request for antenna calibration less frequently in order to conserve battery power.

At action 604, the network unit 105 may transmit an antenna calibration acknowledgement to the UE 115b. In this regard, the network unit 105 may transmit the calibration acknowledgment to the UE 115b via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication. The network unit 105 may respond to the request at action 602 by transmitting an antenna array calibration acknowledgment to the UE 115b.

At action 606, the UE 115b may transmit a plurality of reference signals to the network unit 105. In this regard the UE 115b may transmit a plurality of sounding reference signals (SRSs), demodulation reference signals (DMRSs), phase tracking reference signals (PTRSs), or other suitable reference signals to the network unit 105. In some aspects, the plurality of reference signals is associated with a first set of phase values and gain values to be used with a first antenna array.

At action 608, the network unit 105 may transmit signals to the UE 115. The signals received by the UE 115 from the network unit 105 may be used to determine calibration

coefficients associated with the first set of phase values and gain values for the first antenna array. The calibration coefficients may be used by the UE 115 to increase a signal quality of a transmitted beam for uplink transmissions. Combining gain adjustments with phase adjustments in beamforming may be particularly advantageous in millimeter wave communications to coherently combine energy and overcome high path, propagation, and blockage losses at the higher frequencies of millimeter wave communications. Upon performing gain and phase calibration for beamforming, the antenna elements may operate with more accurate gain and phase settings for a more accurate beam pattern.

At action 610, the UE 115a may transmit calibration coefficients associated with the UE 115a. In some aspects, the network unit 105 may receive calibration coefficients associated with the second set of phase and gain values from the UE 115a and/or one or more additional UEs. For example, the UE 115a may transmit a set of calibration coefficients to the network unit 105. The network unit 105 may use a federated (e.g., crowdsourcing) method of receiving the calibration coefficients from the additional UE(s). The calibration coefficients associated with the additional UEs may be the same or similar (e.g., correlated) to the calibration coefficients associated with the UE 115b. The additional UE(s) may be of a same type and/or class as the UE 115b. For example, the additional UEs may have a same model type as the UE 115b. In some aspects, the additional UEs may have the same components (e.g., amplifiers, phase shifters, mixers, etc.) as the UE 115b.

At action 612, the network unit 105 may transmit the set or a subset of the calibration coefficients received from the UE 115a to the UE 115b.

At action 614, the UE 115 may interpolate a third set of phase/gain calibration coefficients from the first set of phase/gain calibration coefficients and/or the second set of phase/gain calibration coefficients received from the network unit at action 612. The UE 115 may interpolate the third set of phase/gain calibration coefficients from the first set of phase/gain calibration coefficients and/or the second set of phase/gain calibration coefficients using the methods described with reference to FIG. 4.

At action 616, the UE 115 may transmit a communication signal (e.g., UCI, a PUCCH communication, a PUSCH communication, or other suitable uplink communication) to the network unit 105 based on the calibration coefficients associated with the first set of phase and gain values, calibration coefficients associated with the second set of phase and gain values, and/or calibration coefficients associated with the third set of phase and gain values.

FIG. 7 is a block diagram of an exemplary UE 700 according to some aspects of the present disclosure. The UE 700 may be the UE 115 in the network 100, or 200 as discussed above. As shown, the UE 700 may include a processor 702, a memory 704, a joint phase/gain calibration module 708, a transceiver 710 including a modem subsystem 712 and a radio frequency (RF) unit 714, and one or more antennas 716. These elements may be coupled with each other and in direct or indirect communication with each other, for example via one or more buses.

The processor 702 may include a central processing unit (CPU), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a controller, a field programmable gate array (FPGA) device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein. The processor 702 may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a

microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

The memory 704 may include a cache memory (e.g., a cache memory of the processor 702), random access memory (RAM), magnetoresistive RAM (MRAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read only memory (EPROM), electrically erasable programmable read only memory (EEPROM), flash memory, solid state memory device, hard disk drives, other forms of volatile and non-volatile memory, or a combination of different types of memory. In some instances, the memory 704 includes a non-transitory computer-readable medium. The memory 704 may store instructions 706. The instructions 706 may include instructions that, when executed by the processor 702, cause the processor 702 to perform the operations described herein with reference to the UEs 115 in connection with aspects of the present disclosure, for example, aspects of FIGS. 3-7. Instructions 706 may also be referred to as code. The terms "instructions" and "code" should be interpreted broadly to include any type of computer-readable statement(s). For example, the terms "instructions" and "code" may refer to one or more programs, routines, sub-routines, functions, procedures, etc. "Instructions" and "code" may include a single computer-readable statement or many computer-readable statements.

The joint phase/gain calibration module 708 may be implemented via hardware, software, or combinations thereof. For example, the joint phase/gain calibration module 708 may be implemented as a processor, circuit, and/or instructions 706 stored in the memory 704 and executed by the processor 702. In some aspects, the joint phase/gain calibration module 708 may implement the aspects of FIGS. 3-6. For example, the joint phase/gain calibration module 708 may transmit, to a network unit 800, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE 700 for uplink transmissions. The joint phase/gain calibration module 708 may receive, from the network unit 800, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array. The joint phase/gain calibration module 708 may transmit, to the network unit 800, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

As shown, the transceiver 710 may include the modem subsystem 712 and the RF unit 714. The transceiver 710 may be configured to communicate bi-directionally with other devices, such as the BSs 105 and/or the UEs 115. The modem subsystem 712 may be configured to modulate and/or encode the data from the memory 704 and the according to a modulation and coding scheme (MCS), e.g., a low-density parity check (LDPC) coding scheme, a turbo coding scheme, a convolutional coding scheme, a digital beamforming scheme, etc. The RF unit 714 may be configured to process (e.g., perform analog to digital conversion or digital to analog conversion, etc.) modulated/encoded data from the modem subsystem 712 (on outbound transmissions) or of transmissions originating from another source such as a UE 115 or a BS 105. The RF unit 714 may be further configured to perform analog beamforming in con-

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junction with the digital beamforming. Although shown as integrated together in transceiver **710**, the modem subsystem **712** and the RF unit **714** may be separate devices that are coupled together to enable the UE **700** to communicate with other devices.

The RF unit **714** may provide the modulated and/or processed data, e.g. data packets (or, more generally, data messages that may contain one or more data packets and other information), to the antennas **716** for transmission to one or more other devices. The antennas **716** may further receive data messages transmitted from other devices. The antennas **716** may provide the received data messages for processing and/or demodulation at the transceiver **710**. The antennas **716** may include multiple antennas of similar or different designs in order to sustain multiple transmission links. The RF unit **714** may configure the antennas **716**.

In some instances, the UE **700** may include multiple transceivers **710** implementing different RATs (e.g., NR and LTE). In some instances, the UE **700** may include a single transceiver **710** implementing multiple RATs (e.g., NR and LTE). In some instances, the transceiver **710** may include various components, where different combinations of components may implement RATs.

FIG. **8** is a block diagram of an exemplary network unit **800** according to some aspects of the present disclosure. The network unit **800** may be the BS **105**, the CU **210**, the DU **230**, or the RU **240**, as discussed above. As shown, the network unit **800** may include a processor **802**, a memory **804**, a joint phase/gain calibration module **808**, a transceiver **810** including a modem subsystem **812** and a RF unit **814**, and one or more antennas **816**. These elements may be coupled with each other and in direct or indirect communication with each other, for example via one or more buses.

The processor **802** may have various features as a specific-type processor. For example, these may include a CPU, a DSP, an ASIC, a controller, a FPGA device, another hardware device, a firmware device, or any combination thereof configured to perform the operations described herein. The processor **802** may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

The memory **804** may include a cache memory (e.g., a cache memory of the processor **802**), RAM, MRAM, ROM, PROM, EPROM, EEPROM, flash memory, a solid state memory device, one or more hard disk drives, memristor-based arrays, other forms of volatile and non-volatile memory, or a combination of different types of memory. In some instances, the memory **804** may include a non-transitory computer-readable medium. The memory **804** may store instructions **806**. The instructions **806** may include instructions that, when executed by the processor **802**, cause the processor **802** to perform operations described herein, for example, aspects of FIGS. **3-6**. Instructions **806** may also be referred to as code, which may be interpreted broadly to include any type of computer-readable statement(s).

The joint phase/gain calibration module **808** may be implemented via hardware, software, or combinations thereof. For example, the joint phase/gain calibration module **808** may be implemented as a processor, circuit, and/or instructions **806** stored in the memory **804** and executed by the processor **802**.

In some aspects, the joint phase/gain calibration module **808** may implement the aspects of FIGS. **3-6**. For example, the joint phase/gain calibration module **808** the joint phase/gain calibration module **808** may transmit, to UE **700**, a

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plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the network unit **800** for downlink transmissions. The joint phase/gain calibration module **808** may receive, from the UE **700**, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array. The joint phase/gain calibration module **808** may transmit, to the UE **700**, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

Additionally or alternatively, the joint phase/gain calibration module **808** may be implemented in any combination of hardware and software, and may, in some implementations, involve, for example, processor **802**, memory **804**, instructions **806**, transceiver **810**, and/or modem **812**.

As shown, the transceiver **810** may include the modem subsystem **812** and the RF unit **814**. The transceiver **810** may be configured to communicate bi-directionally with other devices, such as the UEs **115** and/or UE **700**. The modem subsystem **812** may be configured to modulate and/or encode data according to a MCS, e.g., a LDPC coding scheme, a turbo coding scheme, a convolutional coding scheme, a digital beamforming scheme, etc. The RF unit **814** may be configured to process (e.g., perform analog to digital conversion or digital to analog conversion, etc.) modulated/encoded data from the modem subsystem **812** (on outbound transmissions) or of transmissions originating from another source such as a UE **115** or UE **700**. The RF unit **814** may be further configured to perform analog beamforming in conjunction with the digital beamforming. Although shown as integrated together in transceiver **810**, the modem subsystem **812** and/or the RF unit **814** may be separate devices that are coupled together at the network unit **800** to enable the network unit **800** to communicate with other devices.

The RF unit **814** may provide the modulated and/or processed data, e.g. data packets (or, more generally, data messages that may contain one or more data packets and other information), to the antennas **816** for transmission to one or more other devices. This may include, for example, a configuration indicating a plurality of sub-slots within a slot according to aspects of the present disclosure. The antennas **816** may further receive data messages transmitted from other devices and provide the received data messages for processing and/or demodulation at the transceiver **810**. The antennas **816** may include multiple antennas of similar or different designs in order to sustain multiple transmission links.

In some instances, the network unit **800** may include multiple transceivers **810** implementing different RATs (e.g., NR and LTE). In some instances, the network unit **800** may include a single transceiver **810** implementing multiple RATs (e.g., NR and LTE). In some instances, the transceiver **810** may include various components, where different combinations of components may implement RATs.

FIG. **9** is a flow diagram of a communication method **900** according to some aspects of the present disclosure. Aspects of the method **900** can be executed by a computing device (e.g., a processor, processing circuit, and/or other suitable component) of a wireless communication device or other suitable means for performing the actions. For example, a wireless communication device, such as the UE **115** or the

UE 700, may utilize one or more components, such as the processor 702, the memory 704, the joint phase/gain module 708, the transceiver 710, the modem 712, and the one or more antennas 716, to execute aspects of method 900. The method 900 may employ similar mechanisms as in the networks 100 and 200 and the aspects and actions described with respect to FIGS. 3-6. As illustrated, the method 900 includes a number of enumerated actions, but the method 900 may include additional actions before, after, and in between the enumerated actions. In some aspects, one or more of the enumerated actions may be omitted or performed in a different order.

At action 910, the method 900 includes a user equipment (UE) (e.g., the UE 115, the UE 700), transmitting a plurality of reference signals to a network unit (e.g., the network unit 800, the BS 105, the RU 240, the DU 230, and/or the CU 210). In this regard the UE may transmit a plurality of sounding reference signals (SRSs), demodulation reference signals (DMRSs), phase tracking reference signals (PTRSs), or other suitable reference signals to the network unit. In some aspects, the plurality of reference signals is associated with a first set of phase values and gain values to be used with a first antenna array. The first antenna array may be an antenna array of one or more antenna arrays used by the UE for uplink transmissions to the network unit. In this regard, the uplink transmissions may include PUCCH transmissions, PUSCH transmissions, PRACH transmissions, and/or other suitable uplink transmissions.

In some aspects, the UE may include one or more antenna arrays at multiple positions in the UE. For example, the antenna arrays may be positioned along different edges (e.g., sides) of the UE for spatial diversity. The antenna arrays may be positioned parallel and/or orthogonal to one another. Each antenna array may include multiple antenna elements. For example, each antenna array may include a single row of antenna elements and/or multiple rows of antenna elements.

At action 920, the method 900 includes the UE receiving signals from the network unit. The signals received by the UE from the network unit may be used to determine calibration coefficients associated with the first set of phase values and gain values for the first antenna array. The calibration coefficients may be used by the UE to increase a signal quality of a transmitted beam for uplink transmissions. For example, signal processing over an array of antenna elements during beamforming may include adjusting a phase of signals corresponding to each of the antenna elements in the array. In some aspects, gain adjustments may be used for improving directionality of beams from the antenna elements, such as to control (e.g., reduce) side lobes and control (e.g., increase) the main lobe. Combining gain adjustments with phase adjustments in beamforming may be particularly advantageous in millimeter wave communications to coherently combine energy and overcome high path, propagation, and blockage losses at the higher frequencies of millimeter wave communications. Upon performing gain and phase calibration for beamforming, the antenna elements may operate with more accurate gain and phase settings for a more accurate beam pattern.

At action 930, the method 900 includes the UE transmitting a communication signal (e.g., UCI, a PUCCH communication, a PUSCH communication, or other suitable uplink communication) to the network unit based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values. In some aspects, the calibration coefficients associated with the second set of phase values

and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values. In this regard, the calibration coefficients associated with the second set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values using a linear interpolation method, a sample and hold interpolation method, a spline interpolation method, or other suitable interpolation method.

In some aspects, the calibration coefficients associated with the second set of phase values and gain values may be interpolated from adjacent calibration coefficients associated with the first set of phase values and gain values. In some aspects, the first and second sets of calibration coefficients may be stored in memory (e.g., memory 704, 804) in the form of a two dimensional table. Referring to FIG. 3, each antenna element 306a to 306n of the antenna may be represented by a different two dimensional table. Each two dimensional table corresponding to each of the antenna elements 306a to 306n may be configured with phase/gain calibration coefficients. The number of columns in the table may be based on the number of phase settings. For example, if the UE includes an x bit phase shifter the number of columns may be 2^x . The number of rows in the table may be based on the number of gain settings. For example, if the UE includes a y bit gain controller the number of rows may be 2^y . In a non-limiting example as shown in FIG. 3, the UE may include a 3 bit phase shifter and a 3 bit gain controller such that the table of calibration coefficients includes 8 rows and 8 columns. Each calibration coefficient in the table may correspond to a row index labeled from G0 to G7 and a column index labeled from P0 to P7. In some aspects, the calibration coefficients associated with the first set of phase values and gain values generated by the over the air calibration process may be represented in a first subset of calibration coefficients in the table. The calibration coefficients associated with the second set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values and stored in a second subset of row/column entries. For example, as shown in FIG. 3, the patterned entries in the table may include calibrated values 302. The unpatterned entries in the table may include interpolated values 304.

In the example of FIG. 3, the interpolated values 304 may be interpolated from one or more calibrated values 302. In a non-limiting example, the interpolated value may be based on one or more of the calibrated values in proximity to interpolated values. In some instances, the calibrated value(s) used to determine the interpolated value may include one or more calibrated values directly adjacent to the interpolated value, one or more calibrated values nearby but spaced from the interpolated value, one or more contiguous calibrated values directly adjacent and/or spaced from the interpolated values, one or more non-contiguous calibrated values, one or more calibrated values surrounding (directly or indirectly) the interpolated value, and/or any other calibrated values. As an example, the interpolated value at G1/P1 may be interpolated from one or more of directly adjacent calibrated values at G1/P0, G2/P1, G1/P2, and/or G0/P1. As another example, the interpolated value at G2/P2 may be interpolated from one or more of calibrated values at G1/P0, G3/P0, G3/P4, and/or G1/P4 that are nearby but spaced from G2/P2. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values surrounding G4/P4, such as G4/P1, G2/P1, G2/P3, G2/P5, G3/P6, G5/P6, G6/P5, G6/P3, etc. In another example, the interpolated value at G4/P4 may be interpolated

lated from one or more calibrated values contiguous to G4/P4, such as G4/P3, G3/P4, G4/P5, and G5/P4. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values non-contiguous to G4/P4, such as G4/P1, G2/P3, G4/P7, and G7/P4. In some aspects, the interpolated values **304** may be interpolated from any number of calibrated values **302** at any position within the table **300a**.

In some aspects, the UE may determine the number of calibrated values **302** and the number of interpolated values **304** in the table **300**. The number of calibrated values **302** and/or the number of interpolated values **304** may be based on a battery power level of the UE, a temperature of the UE, a frequency range associated with the communication signal, or other suitable parameter. In some aspects, the UE may transmit an indicator to the network unit indicating the number of calibrated values **302** and/or the number of interpolated values **304**. In this regard, the UE may transmit the indicator to the network unit via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. Additionally or alternatively, the network unit may transmit an indicator to the UE indicating the number of calibrated values **302** and/or the number of interpolated values **304**. In this regard, the network unit may transmit the indicator to the UE via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication.

In some aspects, the antenna may include n antenna elements **306**. FIG. 3 shows the same configuration for each antenna element **306a** to **306n** in which the table has the same number of calibrated values **302** and interpolated values **304**. However, the present disclosure is not so limited and each antenna element **306a** to **306n** may have a different number of calibrated values **302** and interpolated values **304**.

In some aspects, the UE may transmit a request to the network unit for an antenna array calibration. In this regard, the UE may transmit the calibration request to the network unit via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. The network unit may respond to the request by transmitting an antenna array calibration acknowledgment to the UE. In this regard, the network unit may transmit the calibration acknowledgment to the UE via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication. The UE may transmit the plurality of reference signals to the network unit in response to receiving the antenna array calibration acknowledgement.

In some aspects the phase/gain calibration of the antenna(s) may be affected by certain events associated with the UE. In this regard, the UE may transmit the request for the antenna array calibration based on an event associated with the UE. In some aspects, the event may include a temperature of the UE satisfying a temperature threshold. For example, the temperature sensor of the UE may indicate that the temperature of the UE is greater than, equal to, or less than a temperature threshold. Since the UE temperature may affect the antenna(s) calibration, the UE may transmit the antenna array calibration request based on the temperature of the UE satisfying the temperature threshold.

In some aspects, the event may include a frequency range associated with the communication signal satisfying a frequency threshold. For example, when the UE is configured to switch from a first frequency range (e.g., FR1, FR2, FR3, FR4, FR5) to a second (different) frequency range (e.g., FR1, FR2, FR3, FR4, FR5), the UE may transmit the antenna array calibration request based on the difference in

the first frequency range and the second frequency range satisfying a frequency range threshold.

In some aspects, the UE may transmit the request for the antenna array calibration based on a mobility of the UE. A higher mobility rate of the UE may require more frequent calibration of the antenna elements as compared to a lower mobility rate.

In some aspects, the UE may transmit the request for the antenna array calibration based on a battery power level of the UE. The UE may transmit the request for antenna calibration less frequently when the battery power level is low as compared to when the battery power level is high. Since the antenna calibration process may consume battery power in the UE, the UE may transmit the request for antenna calibration less frequently in order to conserve battery power.

FIG. 10 is a flow diagram of a communication method **1000** according to some aspects of the present disclosure. Aspects of the method **1000** can be executed by a computing device (e.g., a processor, processing circuit, and/or other suitable component) of a wireless communication device or other suitable means for performing the actions. For example, a wireless communication device, such as the UE **115** or the UE **700**, may utilize one or more components, such as the processor **702**, the memory **704**, the joint phase/gain module **708**, the transceiver **710**, the modem **712**, and the one or more antennas **716**, to execute aspects of method **1000**. The method **1000** may employ similar mechanisms as in the networks **100** and **200** and the aspects and actions described with respect to FIGS. 3-6. As illustrated, the method **1000** includes a number of enumerated actions, but the method **1000** may include additional actions before, after, and in between the enumerated actions. In some aspects, one or more of the enumerated actions may be omitted or performed in a different order.

At action **1010**, the method **1000** includes a user equipment (UE) (e.g., the UE **115**, the UE **700**), transmitting a plurality of reference signals to a network unit (e.g., the network unit **800**, the BS **105**, the RU **240**, the DU **230**, and/or the CU **210**). In this regard the UE may transmit a plurality of sounding reference signals (SRSs), demodulation reference signals (DMRSs), phase tracking reference signals (PTRSs), or other suitable reference signals to the network unit. In some aspects, the plurality of reference signals is associated with a first set of phase values and gain values to be used with a first antenna array. The first antenna array may be an antenna array of one or more antenna arrays used by the UE for uplink transmissions to the network unit. In this regard, the uplink transmissions may include PUCCH transmissions, PUSCH transmissions, PRACH transmissions, and/or other suitable uplink transmissions.

In some aspects, the UE may include one or more antenna arrays at multiple positions in the UE. For example, the antenna arrays may be positioned along different edges (e.g., sides) of the UE for spatial diversity. The antenna arrays may be positioned parallel and/or orthogonal to one another. Each antenna array may include multiple antenna elements. For example, each antenna array may include a single row of antenna elements and/or multiple rows of antenna elements.

At action **1020**, the method **900** includes the UE receiving signals from the network unit. The signals received by the UE from the network unit may be used to determine calibration coefficients associated with the first set of phase values and gain values for the first antenna array. The calibration coefficients may be used by the UE to increase a signal quality of a transmitted beam for uplink transmis-

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sions. For example, signal processing over an array of antenna elements during beamforming may include adjusting a phase of signals corresponding to each of the antenna elements in the array. In some aspects, gain adjustments may be used for improving directionality of beams from the antenna elements, such as to control (e.g., reduce) side lobes and control (e.g., increase) the main lobe. Combining gain adjustments with phase adjustments in beamforming may be particularly advantageous in millimeter wave communications to coherently combine energy and overcome high path, propagation, and blockage losses at the higher frequencies of millimeter wave communications. Upon performing gain and phase calibration for beamforming, the antenna elements may operate with more accurate gain and phase settings for a more accurate beam pattern.

At action 1030, the method 1000 includes the UE receiving calibration coefficients associated with a second set of phase and gain values. In this regard, the UE may receive the second set of phase and gain values from the network unit via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication. In some aspects, the network unit may receive the second set of phase and gain values from one or more additional UEs. For example, one or more additional UEs may transmit a set of calibration coefficients to the network unit. The network unit may transmit the set or a subset of the calibration coefficients to the UE. The network unit may use a federated (e.g., crowd-sourcing) method of receiving the calibration coefficients from the additional UE(s). The calibration coefficients associated with the additional UEs may be the same or similar (e.g., correlated) to the calibration coefficients associated with the UE. The additional UE(s) may be of a same type and/or class as the UE. For example, the additional UEs may have a same model type as the UE. In some aspects, the additional UEs may have the same components (e.g., amplifiers, phase shifters, mixers, etc.) as the UE.

At action 1040, the method 1000 includes the UE transmitting a communication signal (e.g., UCI, a PUCCH communication, a PUSCH communication, or other suitable uplink communication) to the network unit based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values. In some aspects, the calibration coefficients associated with the second set of phase values and gain values may be the calibration coefficients associated with the additional UE(s) that was received from the base station. Additionally or alternatively, the UE may transmit the uplink communication using the first set of calibration coefficients, the second set of calibration coefficients and a third set of calibration coefficients that is interpolated from the first set of calibration coefficients. In this regard, the calibration coefficients associated with the third set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set of phase values and gain values using a linear interpolation method, a sample and hold interpolation method, a spline interpolation method, or other suitable interpolation method.

In some aspects, the calibration coefficients associated with the third set of phase values and gain values may be interpolated from adjacent calibration coefficients associated with the first set of phase values and gain values. In some aspects, the first, second, and third sets of calibration coefficients may be stored in memory (e.g., memory 704, 804) in the form of a two dimensional table. Referring to FIG. 4, each antenna element 306a to 306n of the antenna may be represented by a different two dimensional table. Each two dimensional table corresponding to each of the antenna

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elements 306a to 306n may be configured with phase/gain calibration coefficients. The number of columns in the table may be based on the number of phase settings. For example, if the UE includes an x bit phase shifter the number of columns may be 2^x . The number of rows in the table may be based on the number of gain settings. For example, if the UE includes a y bit gain controller the number of rows may be 2^y . In a non-limiting example as shown in FIG. 4, the UE may include a 3 bit phase shifter and a 3 bit gain controller such that the table of calibration coefficients includes 8 rows and 8 columns. Each calibration coefficient in the table may correspond to a row index labeled from G0 to G7 and a column index labeled from P0 to P7. In some aspects, the calibration coefficients associated with the first set of phase values and gain values generated by the over the air calibration process may be represented in a first subset of calibration coefficients in the table. The calibration coefficients associated with the second set of phase and gain values received from the network unit may be represented as a second subset. The calibration coefficients associated with the third set of phase values and gain values may be interpolated from the calibration coefficients associated with the first set and/or the second set and stored in a third subset of row/column entries. For example, as shown in FIG. 4, the cross hatched patterned entries in the table may include calibrated values 302, the unpatterned entries in the table may include interpolated values 304 and the diagonal patterned entries in the table may include the federated values 405.

In the example of FIG. 4, the interpolated values 304 may be interpolated from one or more calibrated values 302 and/or one or more federated values 405. In a non-limiting example, the interpolated values may be based on one or more of the calibrated values 302 and/or one or more federated values 405 in proximity to interpolated values 304. In some instances, the calibrated value(s) used to determine the interpolated value may include one or more calibrated values and/or one or more federated values directly adjacent to the interpolated value, one or more calibrated values and/or one or more federated values nearby but spaced from the interpolated value, one or more contiguous calibrated values and/or one or more contiguous federated values directly adjacent and/or spaced from the interpolated values, one or more non-contiguous calibrated values and/or one or more non-contiguous federated values, one or more calibrated values and/or one or more federated values surrounding (directly or indirectly) the interpolated value, and/or any other calibrated values and/or other federated values. As an example, the interpolated value at G1/P1 may be interpolated from one or more of directly adjacent calibrated values and/or one or more directly adjacent federated values at G1/P0, G2/P1, G1/P2, and/or G0/P1. As another example, the interpolated value at G2/P2 may be interpolated from one or more of calibrated values and/or one or more federated values at G1/P0, G3/P0, G3/P4, and/or G1/P4 that are nearby but spaced from G2/P2. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values and/or one or more federated values surrounding G4/P4, such as G4/P1, G2/P1, G2/P3, G2/P5, G3/P6, G5/P6, G6/P5, G6/P3, etc. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values and/or one or more federated values contiguous to G4/P4, such as G4/P3, G3/P4, G4/P5, and G5/P4. In another example, the interpolated value at G4/P4 may be interpolated from one or more calibrated values and/or one or more federated values non-contiguous to G4/P4, such as G4/P1, G2/P3, G4/P7, and G7/P4. In some

aspects, the interpolated values **304** may be interpolated from any number of calibrated values **302** and/or any number of federated values at any position within the table **300a**.

In some aspects, the UE may determine the number of calibrated values **302**, the number of interpolated values **304**, and/or the number of federated values **405** in the table **300**. The number of calibrated values **302**, the number of interpolated values **304**, and/or the number of federated values **405** may be based on a battery power level of the UE, a temperature of the UE, a frequency range associated with the communication signal, or other suitable parameter. In some aspects, the UE may transmit an indicator to the network unit indicating the number of calibrated values **302**, the number of interpolated values **304**, and/or the number of federated values **405**. In this regard, the UE may transmit the indicator to the network unit via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. Additionally or alternatively, the network unit may transmit an indicator to the UE indicating the number of calibrated values **302**, the number of interpolated values **304**, and/or the number of federated values **405**. In this regard, the network unit may transmit the indicator to the UE via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication.

In some aspects, the antenna may include n antenna elements **306**. FIG. 4 shows the same configuration for each antenna element **306a** to **306n** in which each table has the same number of calibrated values **302** and interpolated values **304**. However, the present disclosure is not so limited and each antenna element **306a** to **306n** may have a different number of calibrated values **302**, interpolated values **304**, and federated values **405**.

In some aspects, the UE may transmit a request to the network unit for an antenna array calibration. In this regard, the UE may transmit the calibration request to the network unit via UCI, a PUCCH communication, a PUSCH communication and/or other suitable communication. The network unit may respond to the request by transmitting an antenna array calibration acknowledgment to the UE. In this regard, the network unit may transmit the calibration acknowledgment to the UE via DCI, a PDCCH communication, a PDSCH communication and/or other suitable communication. The UE may transmit the plurality of reference signals to the network unit in response to receiving the antenna array calibration acknowledgement.

In some aspects the phase/gain calibration of the antenna(s) may be affected by certain events associated with the UE. In this regard, the UE may transmit the request for the antenna array calibration based on an event associated with the UE. In some aspects, the event may include a temperature of the UE satisfying a temperature threshold. For example, the temperature sensor of the UE may indicate that the temperature of the UE is greater than, equal to, or less than a temperature threshold. Since the UE temperature may affect the antenna(s) calibration, the UE may transmit the antenna array calibration request based on the temperature of the UE satisfying the temperature threshold.

In some aspects, the event may include a frequency range associated with the communication signal satisfying a frequency threshold. For example, when the UE is configured to switch from a first frequency range (e.g., FR1, FR2, FR3, FR4, FR5) to a second (different) frequency range (e.g., FR1, FR2, FR3, FR4, FR5), the UE may transmit the antenna array calibration request based on the difference in the first frequency range and the second frequency range satisfying a frequency range threshold.

In some aspects, the UE may transmit the request for the antenna array calibration based on a mobility of the UE. A higher mobility rate of the UE may require more frequent calibration of the antenna elements as compared to a lower mobility rate.

In some aspects, the UE may transmit the request for the antenna array calibration based on a battery power level of the UE. The UE may transmit the request for antenna calibration less frequently when the battery power level is low as compared to when the battery power level is high. Since the antenna calibration process may consume battery power in the UE, the UE may transmit the request for antenna calibration less frequently in order to conserve battery power.

Further aspects of the present disclosure include the following:

Aspect 1 includes a method of wireless communication performed by a user equipment (UE), the method comprising transmitting, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions; receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; and transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

Aspect 2 includes the method of aspect 1, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from adjacent calibration coefficients associated with the first set of phase and gain values.

Aspect 3 includes the method of any of aspects 1-2, wherein the first set of phase and gain values and the second set of phase and gain values are associated with a same antenna element.

Aspect 4 includes the method of any of aspects 1-3, further comprising transmitting, to the network unit, a request for an antenna array calibration; and receiving, from the network unit, an antenna array calibration acknowledgement, wherein the transmitting the plurality of reference signals comprises transmitting the plurality of reference signals in response to the antenna array calibration acknowledgement.

Aspect 5 includes the method of any of aspects 1-4, wherein the transmitting the request for the antenna array calibration is based on an event associated with the UE.

Aspect 6 includes the method of any of aspects 1-5, wherein the event comprises at least one of a temperature of the UE satisfying a temperature threshold; a frequency range associated with the communication signal satisfying a frequency threshold; or a distance between the UE and the network unit satisfying a distance threshold.

Aspect 7 includes the method of any of aspects 1-6, further comprising receiving, from the network unit, calibration coefficients associated with a third set of phase and gain values, wherein the transmitting the communication signal further comprises transmitting the communication signal further based on the third set of phase and gain values.

Aspect 8 includes a method of wireless communication performed by a user equipment (UE), the method comprising transmitting, to a network unit, a plurality of reference

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signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions; receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; receiving, from the network unit, signals used to determine the calibration coefficients associated with a second set of phase and gain values; and transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values.

Aspect 9 includes the method of aspect 8, wherein the signals used to determine the calibration coefficients associated with the second set of phase and gain values comprise signals from one or more additional UEs.

Aspect 10 includes the method of any of aspects 8-9, wherein the one or more additional UEs are from a same UE class as the UE.

Aspect 11 includes the method of any of aspects 8-10, further comprising: transmitting, to the network unit, a request for an antenna array calibration; and

receiving, from the network unit, an antenna array calibration acknowledgement, wherein the transmitting the plurality of reference signals comprises transmitting the plurality of reference signals in response to the antenna array calibration acknowledgement.

Aspect 12 includes the method of any of aspects 8-11, wherein the transmitting the request for the antenna array calibration is based on an event associated with the UE.

Aspect 13 includes the method of any of aspects 8-12, wherein the event comprises at least one of a temperature of the UE satisfying a temperature threshold; a frequency range associated with the communication signal satisfying a frequency threshold; or a distance between the UE and the network unit satisfying a distance threshold.

Aspect 14 includes a non-transitory computer-readable medium storing one or more instructions for wireless communication, the one or more instructions comprising one or more instructions that, when executed by one or more processors of a first wireless communication device perform any one of aspects 1-7.

Aspect 15 includes a non-transitory computer-readable medium storing one or more instructions for wireless communication, the one or more instructions comprising one or more instructions that, when executed by one or more processors of a first wireless communication device, cause the network unit to perform any one of aspects 8-13.

Aspect 16 includes a first wireless communication device comprising one or more means to perform any one or more of aspects 1-7.

Aspect 17 includes a first wireless communication device comprising one or more means to perform any one or more of aspects 8-13.

Aspect 18 includes a first wireless communication device comprising a memory; a transceiver; and at least one processor coupled to the memory and the transceiver, wherein the first wireless communication device is configured to perform any one or more of aspects 1-7.

Aspect 19 includes a first wireless communication device comprising a memory; a transceiver; and at least one processor coupled to the memory and the transceiver, wherein the first wireless communication device is configured to perform any one or more of aspects 8-13.

Information and signals may be represented using any of a variety of different technologies and techniques. For

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example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described above may be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, "or" as used in a list of items (for example, a list of items prefaced by a phrase such as "at least one of" or "one or more of") indicates an inclusive list such that, for example, a list of [at least one of A, B, or C] means A or B or C or AB or AC or BC or ABC (i.e., A and B and C).

As those of some skill in this art will by now appreciate and depending on the particular application at hand, many modifications, substitutions and variations may be made in and to the materials, apparatus, configurations and methods of use of the devices of the present disclosure without departing from the spirit and scope thereof. In light of this, the scope of the present disclosure should not be limited to that of the particular instances illustrated and described herein, as they are merely by way of some examples thereof, but rather, should be fully commensurate with that of the claims appended hereafter and their functional equivalents.

What is claimed is:

1. A method of wireless communication performed by a user equipment (UE), the method comprising:

transmitting, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions;

receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; and transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients asso-

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ciated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

2. The method of claim 1, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from adjacent calibration coefficients associated with the first set of phase and gain values.

3. The method of claim 1, wherein the first set of phase and gain values and the second set of phase and gain values are associated with a same antenna element.

4. The method of claim 1, further comprising:
transmitting, to the network unit, a request for an antenna array calibration; and
receiving, from the network unit, an antenna array calibration acknowledgement, wherein the transmitting the plurality of reference signals comprises transmitting the plurality of reference signals in response to the antenna array calibration acknowledgement.

5. The method of claim 4, wherein the transmitting the request for the antenna array calibration is based on an event associated with the UE.

6. The method of claim 5, wherein the event comprises at least one of:

a temperature of the UE satisfying a temperature threshold;
a frequency range associated with the communication signal satisfying a frequency threshold; or
a distance between the UE and the network unit satisfying a distance threshold.

7. The method of claim 1, further comprising:
receiving, from the network unit, calibration coefficients associated with a third set of phase and gain values, wherein the transmitting the communication signal comprises transmitting the communication signal further based on the third set of phase and gain values.

8. A method of wireless communication performed by a user equipment (UE), the method comprising:

transmitting, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions;

receiving, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array;

receiving, from the network unit, signals used to determine the calibration coefficients associated with a second set of phase and gain values; and

transmitting, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values.

9. The method of claim 8, wherein the signals used to determine the calibration coefficients associated with the second set of phase and gain values comprise signals from one or more additional UEs.

10. The method of claim 9, wherein the one or more additional UEs are from a same UE class as the UE.

11. The method of claim 8, further comprising: transmitting, to the network unit, a request for an antenna array calibration; and

receiving, from the network unit, an antenna array calibration acknowledgement, wherein the transmitting the plurality of reference signals comprises transmitting the plurality of reference signals in response to the antenna array calibration acknowledgement.

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12. The method of claim 11, wherein the transmitting the request for the antenna array calibration is based on an event associated with the UE.

13. The method of claim 12, wherein the event comprises at least one of:

a temperature of the UE satisfying a temperature threshold;
a frequency range associated with the communication signal satisfying a frequency threshold; or
a distance between the UE and the network unit satisfying a distance threshold.

14. A user equipment (UE) comprising:

a memory;
a transceiver; and

at least one processor coupled to the memory and the transceiver, wherein the UE is configured to:

transmit, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions;

receive, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array; and

transmit, to the network unit, a communication signal based on the calibration coefficients associated with the first set of phase and gain values and calibration coefficients associated with a second set of phase and gain values, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from the calibration coefficients associated with the first set of phase and gain values.

15. The UE of claim 14, wherein the calibration coefficients associated with the second set of phase and gain values are interpolated from adjacent calibration coefficients associated with the first set of phase and gain values.

16. The UE of claim 14, wherein the first set of phase and gain values and the second set of phase and gain values are associated with a same antenna element.

17. The UE of claim 14, wherein the calibration coefficients associated with the second array of phase values and gain values is linearly interpolated from the calibration coefficients associated with the first array of phase values and gain values.

18. The UE of claim 14, wherein the calibration coefficients associated with the second array of phase values and gain values is interpolated from the calibration coefficients associated with the first array of phase values and gain values using a sample and hold interpolation.

19. The UE of claim 14, wherein the calibration coefficients associated with the second array of phase values and gain values is interpolated from the calibration coefficients associated with the first array of phase values and gain values using a spline interpolation.

20. The UE of claim 14, wherein the UE is further configured to:

transmit, to the network unit, a request for an antenna array calibration;

receive, from the network unit, an antenna array calibration acknowledgement; and

transmit the plurality of reference signals in response to the antenna array calibration acknowledgement.

21. The UE of claim 20, wherein the UE is further configured to transmit the request for the antenna array calibration based on an event associated with the UE.

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22. The UE of claim 21, wherein the event comprises at least one of:

- a temperature of the UE satisfying a temperature threshold;
- a frequency range associated with the communication signal satisfying a frequency threshold; or
- a distance between the UE and the network unit satisfying a distance threshold.

23. The UE of claim 14, wherein the UE is further configured to:

- receive, from the network unit, calibration coefficients associated with a third set of phase and gain values, wherein the transmitting the communication signal comprises transmitting the communication signal further based on the third set of phase and gain values.

24. A user equipment (UE) comprising:

a memory;

a transceiver; and

at least one processor coupled to the memory and the transceiver, wherein the UE is configured to:

transmit, to a network unit, a plurality of reference signals, wherein the plurality of reference signals is associated with a first set of phase and gain values to be used with a first antenna array at the UE for uplink transmissions;

receive, from the network unit, signals used to determine calibration coefficients associated with the first set of phase and gain values for the first antenna array;

receive, from the network unit, signals used to determine the calibration coefficients associated with a second set of phase and gain values; and

transmit, to the network unit, a communication signal based on the calibration coefficients associated with the

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first set of phase and gain values and the calibration coefficients associated with the second set of phase and gain values.

25. The UE of claim 24, wherein the signals used to determine the calibration coefficients associated with the second set of phase and gain values comprise signals from one or more additional UEs.

26. The UE of claim 25, wherein the one or more additional UEs are from a same UE class as the UE.

27. The UE of claim 24, wherein the UE is further configured to:

transmit, to the network unit, a request for an antenna array calibration;

receive, from the network unit, an antenna array calibration acknowledgement; and

transmit the plurality of reference signals in response to the antenna array calibration acknowledgement.

28. The UE of claim 24, wherein the UE is further configured to transmit the request for the antenna array calibration based on an event associated with the UE.

29. The UE of claim 28, wherein the event comprises at least one of:

a temperature of the UE satisfying a temperature threshold;

a frequency range associated with the communication signal satisfying a frequency threshold; or

a distance between the UE and the network unit satisfying a distance threshold.

30. The UE of claim 24, wherein the UE is further configured to transmit the communication signal further based on calibration coefficients associated with a third set of phase and gain values stored in a lookup table of the UE.

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